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**DEPARTMENT OF STATISTICS
(UGC-SAP-DRS-I ASSISTED) 2024-2025**



PROJECT REPORT ON
**“ STATISTICAL STUDY OF POLITICAL AWARENESS AND
ENTRANCE POLL ”**

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CERTIFICATE

This is to certify that,

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Have Successfully Completed the project entitled,

**“ STATISTICAL STUDY OF POLITICAL AWARENESS AND
ENTRANCE POLL ”**

**Submitted to the Department of Statistics, Dr. Babasaheb
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1. INTRODUCTION

1) Introduction.

Politics is a part of our daily life. It refers to the activities, actions, and policies that are used to gain and hold power in a government. In simple words, politics is the process of making decisions that apply to all members of a society. It involves the creation of laws, rules, and policies that shape how a country is run.

Every country has different political parties and leaders who compete in elections to gain the trust and support of the people. These parties have different ideas, values, and plans for the development of the country. Citizens vote in elections to choose the party or leader they believe will best represent their interests.

Awareness means having knowledge or understanding about something. When we say someone is "aware," we mean that the person understands or is informed about a particular topic or issue.

Political awareness means how much a person knows and understands about politics. This includes knowledge of political parties, leaders, government functions, election processes, current events, and citizens' rights and duties. A politically aware citizen is someone who knows:

- What the different political parties stand for,
- Who the candidates are,
- What the main issues in an election are,
- How the government works,
- And how to vote responsibly.

Political awareness plays an important role in a democracy. When people are politically aware, they can make better decisions during elections. They can choose leaders based on performance and policies rather than on emotions or influence. It helps create a responsible and active citizen.

An entrance poll is a survey conducted just before voters enter a polling place to cast their votes. It is used to gather information about:

- Who voters are voting for
- Why did they make their choice
- Demographic details (such as age, gender, education, etc.)

Key features of entrance polls:

- Conducted before the vote is cast (unlike exit polls, which are done after)
- May help analysts understand voter behavior and preferences on Election Day

- Sometimes used to predict early results or understand voter motivations

Politics plays a vital role in the functioning of any democratic society. It refers to the process of decision-making by which a group of people, often through elected representatives, governs a country. Political systems include the formation of government, law-making, policy development, and public administration. In democratic nations, the power to choose political representatives lies with the people, which makes public involvement essential.

For the democratic process to function effectively, citizens must be politically aware. Political awareness refers to the level of understanding individuals have about the political system, political parties, leaders, election processes, current issues, and their rights and responsibilities as voters. A politically aware voter is more likely to make informed decisions, critically evaluate party policies, and participate actively in democratic activities beyond just casting a vote.

In recent years, political awareness among the public has gained significant attention, especially during elections. One way to assess this awareness and understand voting behavior is through entrance polls. An entrance poll is a survey conducted just after a person has voted, gathering immediate responses about whom they voted for and why. It also collects information about the voter's background, such as age, gender, education, occupation, and income. Entrance polls are useful in analyzing voter behavior and understanding the influence of various socio-demographic factors on political preference.

The present study, titled "Statistical Study of Political Awareness and Entrance Poll," focuses on examining the level of political awareness among voters and analyzing the patterns of party support using entrance poll data. The study aims to explore how demographic factors such as age, gender, education level, income group, and occupation influence political awareness and voting choices.

To carry out this analysis, statistical methods are used to organize and interpret the collected data. In particular, multinomial logistic regression is applied to identify which variables significantly affect a person's choice of political party. This method is suitable when the outcome variable (in this case, political party preference) has more than two categories. It helps in estimating the probability of a voter supporting a particular party based on their background characteristics.

This research contributes to a better understanding of the factors that shape political opinions and behavior. It provides insights into how different groups in society perceive and engage with the political process. The findings of this study can be useful for policymakers, political parties, educators, and social organizations in planning voter awareness campaigns and strengthening democratic participation. Overall, this project emphasizes the importance of political knowledge and informed voting in building a responsible and vibrant democracy.

1.1) Legislative System

I. Lok Sabha

The Lok Sabha is also called the House of the People. It is the lower house of Parliament. Members of the Lok Sabha are chosen by people who are 18 years and older. These members are elected for five years. The Lok Sabha meets in the Parliament House in New Delhi. The total number of seats can go up to 552, but currently, there are 543 elected members. The new Parliament has 888 seats for the Lok Sabha. Out of 543 seats, 131 are reserved for Scheduled Castes and Scheduled Tribes. If needed, the Lok Sabha can be dissolved earlier or its term extended during emergencies.

II. Rajya Sabha

The Rajya Sabha is the upper house of Parliament and is also called the Council of States. It can have a maximum of 250 members. Out of these, 238 are elected by state legislatures and 12 are nominated by the President. These members serve for six years, but every two years, one-third of the members retire. The Rajya Sabha is a permanent body and is not dissolved like the Lok Sabha. It has almost equal power as the Lok Sabha in making laws. However, in money matters, the Lok Sabha has more power. If there is a disagreement, both houses meet together, and the Lok Sabha usually has the final say because it has more members. The Vice President of India is the chairman of the Rajya Sabha.

III. State Governments

India has 28 states and 8 union territories. Each state has its own government. The leader of a state government is the Chief Minister. States have a law-making body called the Legislative Assembly or Vidhan Sabha. Some states also have a second house called the Legislative Council or Vidhan Parishad. States with one house have a "unicameral" legislature, and states with two houses have a "bicameral" legislature. The Vidhan Sabha works like the Lok Sabha, and the Vidhan Sabha works like the Rajya Sabha.

IV. Vidhan Sabha

The Vidhan Sabha is the State Legislative Assembly. It is the main law-making body in the states. Members are called MLAs (Members of the Legislative Assembly). They are elected for five years by the people in their area. There must be at least 60 members and no more than 500, but small states like Goa and Sikkim have fewer members. The Vidhan Sabha can be dissolved early if the government loses its majority or in an emergency.

1.2) Election Commission of India

The Election Commission of India (ECI) is an independent body formed under the Constitution. It looks after and controls the election process in both central and state governments. The main job of the ECI is to make sure that all elections are free and fair. It gets its powers from Article 324 of the Indian Constitution. This Commission is a permanent constitutional body and is responsible for conducting elections for the Parliament, State Legislative Assemblies, and also for the posts of the President and Vice President of India. The Commission also checks the behavior of political parties and leaders before, during, and after elections, making sure they follow the law. If there is any issue or confusion during elections and the law does not clearly explain what to do, the Election Commission has the power to make the right decision. The first Chief Election Commissioner of India was Sukumar Sen. The ECI conducts elections for the President, Vice President, Lok Sabha, Rajya Sabha, and State Assemblies, including union territories like Delhi, Jammu & Kashmir, and Puducherry.

I. State Election Commissions

Apart from the Central Election Commission, every state has its own State Election Commission (SEC). This body looks after the elections for the local governments, such as Panchayats in villages and Municipalities in cities. These local bodies form the third level of government in India. The State Election Commission is also a constitutional body and has full control over conducting elections to these local institutions. The State Election Commissioner is appointed by the Governor of the state and holds office for 5 years. He cannot be removed easily, except in the same way as a High Court judge. All local body elections are managed by the State Election Commission to ensure fairness at the local level.

II. Government of Maharashtra

Maharashtra is the third-largest state in India by area, with over 112 million people. Its main city is Mumbai, which has around 18 million people. Nagpur is the second capital, used in winter. Maharashtra's government follows the parliamentary system. Power is given to city councils, district councils (zilla parishad), subdistrict councils (taluka), and village councils (gram panchayat). A large number of political leaders come from the Maratha–Kunbi community, which plays a big role in the state's politics.

In the early years, the Congress party was the most powerful. But over time, parties like Shiv Sena and BJP became stronger. Many politicians in Maharashtra belong to political families and sometimes change parties to stay in power. In recent years, many alliances have broken and new ones formed. For example, Shiv Sena and BJP separated, and then Congress, NCP, and Shiv Sena came together. Later, a group of Shiv Sena leaders joined the BJP and formed a new government.

III. Structure of the State Government

Maharashtra has two legislative houses: the Legislative Assembly (Vidhan Sabha) and the Legislative Council (Vidhan Parishad). The Assembly has 288 members, elected for five years. Some seats are reserved for Scheduled Castes and Tribes. The Council has

78 members chosen through different groups like local bodies, graduates, teachers, and nominations by the Governor.

The Chief Minister is the head of the government and leads the ministers. The Governor is the formal head, chosen by the President of India for five years.

1.3) Maharashtra in the Indian Parliament

Maharashtra sends members to both houses of the Indian Parliament. It elects 48 members to the Lok Sabha (lower house) and 19 members to the Rajya Sabha (upper house). Lok Sabha members are directly elected by the public, while Rajya Sabha members are elected by the state assembly.

I. Major Political Parties and Alliances in Maharashtra

For many years, the Congress party ruled Maharashtra. Important leaders like Yashwantrao Chavan and Sharad Pawar came from this party. In 1999, Pawar started the Nationalist Congress Party (NCP) after splitting from Congress, but still worked with Congress in elections.

In 1995, Shiv Sena and BJP won the elections and formed a coalition government. For many years, Congress and NCP stayed together while Shiv Sena and BJP were in another alliance. But in 2014, disagreements led to new arrangements. BJP became the largest party, and Shiv Sena joined later.

In 2019, the BJP and Shiv Sena won many seats but couldn't agree on forming a government. So, Shiv Sena joined hands with NCP and Congress. However, in 2022, many Shiv Sena members left to join the BJP again, led by Eknath Shinde.

Other parties in the state include MNS, CPI, SP, RPI, and BSP.

II. Political Power and Communities

The Maratha–Kunbi community, which makes up 31% of the population, holds a lot of political power in Maharashtra. They control many cooperative banks, sugar factories, and political posts. Many chief ministers have been from this group.

Shiv Sena, started by Bal Thackeray, was created to support Marathi people. Over time, it gained more power, especially in Mumbai. BJP, linked to RSS, first got support from urban Brahmins and other upper castes, but later expanded to Marathas and OBCs. Since 2014, the BJP has become the most powerful party in the state.

In 2006, Raj Thackeray left Shiv Sena and formed Maharashtra Navnirman Sena (MNS), focusing on Marathi interests and speaking against migrants.

The Neo-Buddhist (Mahar) community is the second largest group and supports various RPI parties. 25 assembly seats are reserved for Scheduled Castes, and parties often select candidates carefully to win these seats.

III. 2024 Vidhan Sabha Elections

Although the INDIA alliance (Congress, NCP, Shiv Sena UBT) won big in the 2024 Lok Sabha elections, they lost badly in the 2024 Vidhan Sabha elections. The Mahayuti alliance (BJP, Shinde's Shiv Sena, and Ajit Pawar's NCP) won about 80% of the seats. Devendra Fadnavis became the Chief Minister with Shinde and Ajit Pawar as Deputy Chief Ministers. Analysts believe the Ladki Bahin Yojana (which gave money to

women), strong RSS campaign efforts, Narendra Modi's popularity, and Hindu unity slogans helped them win.

1.4. REVIEW OF LITERATURE

1.4.1: Title: The Effect of College on Political Awareness and Knowledge

Author: Harvey E. Rich.

Published in: With & Society, Volume 8, Issue 1, September 1976.

Publisher: Sage Publications

Introduction: This paper explores how college education influences political awareness and understanding of democracy. It highlights that courses like social sciences and liberal arts may boost political interest, knowledge, and participation among students

Objectives:

1. To see how college affects political awareness, knowledge, and interest.
2. To compare political awareness between college students and non-college weth.
3. To find out which parts of college education help students get involved in politics.

Analysis Tools:

1. Pearson correlation and regression analysis to study links between education and political awareness.
2. Z-scores to compare college vs. non-college groups.

Statistical Analysis:

1. Correlations: Show a link between time in college and political awareness.
2. Regression: Highlights how specific courses (like social or political studies) influence awareness.
3. Comparative Analysis: Compares political understanding between college and non-college individuals.

Conclusion: College does help improve political awareness, especially through social and political courses. However, education mainly focuses on job training and often overlooks promoting democratic values.

1.4.2: Title: Political Awareness and Its Implications on Participatory Behaviour: A Study of Naga Women Voters In Nagaland

Author: MOAMENLA AMER

Published in: Indian Journal of Gender Studies, Volume 16, Issue 3, 2009, Pages 359-374

Publisher: SAGE Publications

Introduction: This study explores how aware Naga women are about politics and how that affects their voting. In Nagaland, traditional and male-dominated norms often limit women's political roles. The study also looks at how media and political discussions influence their awareness and voting behaviour.

Objectives

1. To measure Naga women's political knowledge.
2. To see how media and discussions affect their awareness.
3. To check if awareness leads to higher election participation.

Hypotheses

1. Politically aware women are more active in elections.
2. News and political talks help increase women's political awareness.

Statistical Analysis and Tools

1. Used descriptive stats, cross-tabulation, and comparisons to find patterns.
2. Key factors studied: age, education, marital status, job, place of living, and how often they discuss politics.
3. Focused on election participation, media use, and awareness levels.

Conclusion

1. Naga women had moderate political awareness, but it didn't strongly affect voting, which is more shaped by culture and politics.
2. Women are more active in social roles like peace efforts, not formal politics.
3. Patriarchal systems still limit their chances in political spaces.

1.4.3 Title: Political Awareness, Knowledge, and Participation Relationship Using Structural Equation Modeling Approach

Authors: Aidinil Zetra, Kartini Aboo Talib @Khalid, Ferra Yanuar, and Susi Marisa

Journal: Jurnal Wacana Politik, Vol. 7, No. 1, March 2022

DOI: 10.24198/jwp.v7i1.36325

Introduction

This study explores how political awareness and knowledge affect participation in Padang, Indonesia, using Structural Equation Modeling (SEM). It highlights participation as essential for democracy and looks at how knowledge acts as a link between awareness and action.

Objectives

1. To see how awareness affects political knowledge.
2. To check if knowledge connects awareness and participation.
3. To build a model explaining political participation using SEM.

Hypotheses

1. H1: Awareness boosts political knowledge.
2. H2: Awareness indirectly affects participation.
3. H3: Knowledge improves participation.

Analysis Tools

1. SEM to build the model.
2. Bootstrap method to test reliability.
3. Goodness-of-fit checks (RMSEA, CFI, TLI) to validate the model.

Statistical Analysis and Tools

1. Cronbach's alpha > 0.9 showed high reliability.
2. Robust SEM method used for small samples.
3. Bootstrap confirmed accurate estimates.

Conclusion

Political knowledge plays a key role in turning awareness into participation. The SEM model is reliable, but future research should include factors like social media and political trust for better insights.

1.4.4 Title: Women, Political Participation, and the Arab Spring: Awareness and Participation in Tunisia

Author: Carla B. Abdo-Katsipis

Introduction:

The Arab Spring created new political spaces. This study focuses on how women in Tunisia have been affected, whether they've gained opportunities or remained marginalized.

Objectives:

1. Measure political awareness in men and women.
2. Identify barriers to women's participation.
3. Study the effects of education, income, and access to information.
4. Explore regional/cultural factors.
5. Recommend ways to boost female political involvement.

Hypotheses:

1. Women are less politically aware due to structural and cultural barriers.
2. Education and urban living increase awareness, especially for women.
3. Income influences awareness more for men than women.

Methods & Tools:

1. Data from a 2011 voter survey (1,506 people).
2. Logistic regression to analyze variables like gender, income, education, and region.
3. Political awareness measured by ability to identify the 2011 election type.

Key Findings:

1. Women are 40% less aware than men.
2. Education boosts awareness more for women than men.
3. Urban women are more informed than rural ones.
4. Information access makes women nearly 4x more aware.
5. Cultural norms still restrict women's political roles.

Conclusion:

Women face major barriers—lack of education, poor access to information, and cultural limits. Political awareness is key to participation. Gender equity needs targeted efforts.

1.5. OBJECTIVES

1. To compare education-wise differences in political awareness.
2. To analyze gender-wise variation in political awareness.
3. To examine the influence of media consumption on political awareness.
4. To identify the impact of geographic location on political awareness.
5. To evaluate how income levels affect political awareness.

1.6. DATA AND METHODOLOGY

6.1. Study Area

Chhatrapati Sambhaji Nagar, formerly known as Aurangabad, is a major city in the Marathwada region of Maharashtra, India. It holds cultural, historical, and political significance in the state. The city is a part of the Aurangabad district and plays an active role in state and national elections.

Chhatrapati Sambhaji Nagar district currently comprises six Vidhan Sabha (Assembly) constituencies, which are:

1. Aurangabad Central
2. Aurangabad East
3. Aurangabad West (SC)
4. Vaijapur
5. Gangapur
6. Kannad



All six constituencies fall under the Aurangabad Lok Sabha (Parliamentary) Constituency. For this project, the Aurangabad Central Assembly Constituency is selected as the focus area. It is a key urban constituency that reflects a mix of residential, commercial, and semi-urban voters. The constituency has approximately 3.25 lakh registered voters, representing a wide range of social, educational, occupational, and economic backgrounds. This location is ideal for studying political awareness and voting behavior, as it includes a balanced population mix and active electoral participation.

6.2. Sampling Method

The simple random sampling method is used for data collection. In this method, each individual in the voter population has an equal and independent chance of being selected. This reduces bias and ensures a more representative and reliable sample for statistical analysis.

Respondents are selected from different wards and neighborhoods within the Aurangabad Central constituency. Care is taken to include people from diverse groups based on:

1. Gender
2. Age
3. Education level
4. Caste
5. Income group
6. Locality (urban vs. semi-urban)

6.3.Data Collection Process:

A structured questionnaire is prepared to collect data. The questionnaire includes:

1. Demographic questions (age, gender, caste, education, income, occupation)
2. Political awareness questions (knowledge about parties, leaders, policies, and current issues)
3. Media consumption behavior (frequency and source of political news)
4. Opinions on political factors (e.g., influence of specific community leaders or issues)
5. Voting preferences (based on party or alliance)

6.4. Data is collected through:

1. Face-to-face interviews
2. Printed survey forms
3. Online Google Forms

Efforts are made to ensure fair representation across groups to allow meaningful analysis of how awareness and voting patterns vary among different sections of society.

6.5. Tools Used for Analysis

The collected data is analyzed using various statistical tools, including:

1. Chi-Square Test of Independence – to test relationships between demographic variables and awareness
2. t-tests and Kruskal-Wallis tests – to compare awareness scores between groups
3. Multiple Linear Regression – to identify predictors of political awareness
4. Multinomial Logistic Regression – to examine the influence of background variables on political party preference
5. These tools help in identifying key patterns and drawing valid, evidence-based conclusions about the state of political awareness in Aurangabad Central.

2. Methods & Research Methodology

2.1. CHI-SQUARE TEST:

The Chi-squared test of independence is used to determine if there is a significant relationship between two nominal (categorical) variables. The frequency of each category for one nominal variable is compared across the categories of the second nominal variable. The data can be displayed in a contingency table where each row represents a category for one variable and each column represents a category for the other variable. For example, say a researcher wants to examine the relationship between gender (male vs. female) and empathy (high vs. low). The chi-square test of independence can be used to examine this relationship

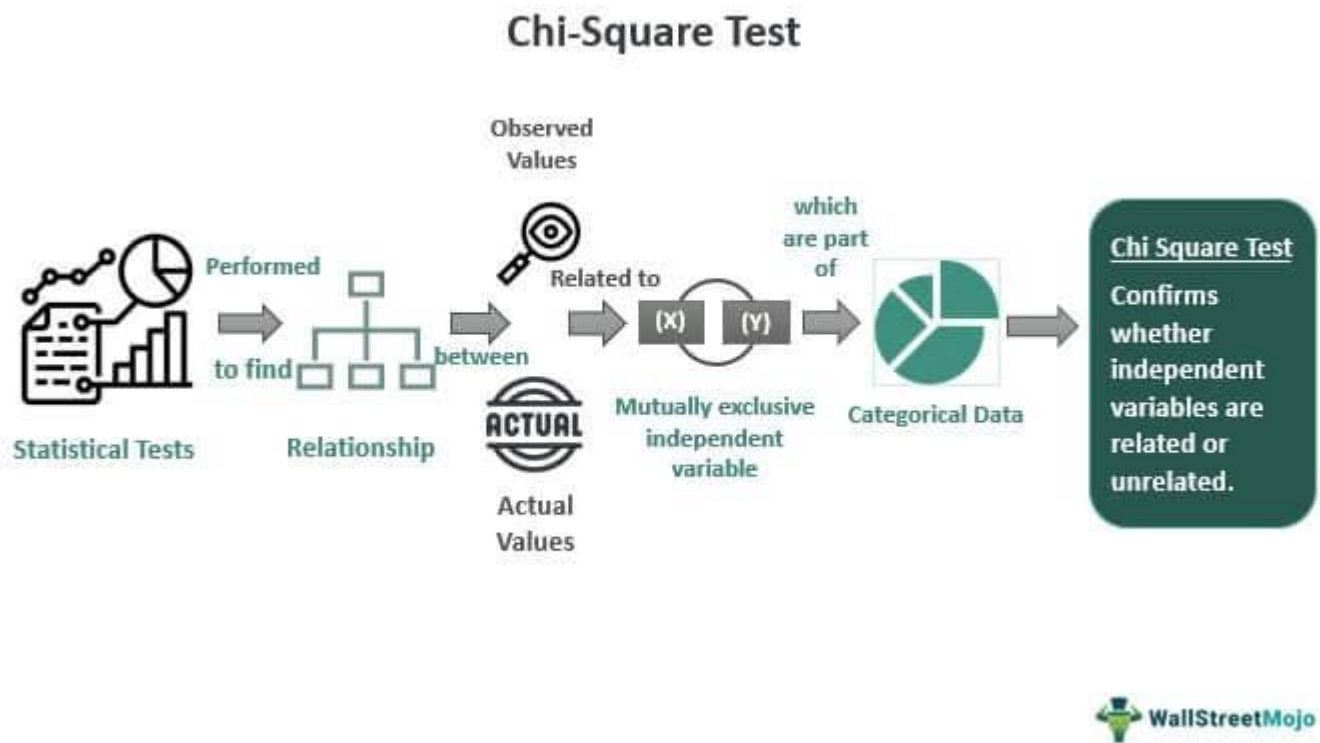


Chart No.2.1

How to calculate the test statistic:

Pearson's chi-square (X^2) is the test statistic for the chi-square test of independence:

$$X^2 = \sum \frac{(O - E)^2}{E}$$

- X^2 is the chi-square test statistic
- Σ is the summation operator (it means "take the sum of")
- O is the observed frequency
- E is the expected frequency

The chi-square test statistic measures how much the observed frequencies differ from the frequencies we would expect if the two variables are unrelated. It is large when there's a big difference between the observed and expected frequencies ($O-E$ in the equation).

Hypothesis:

H_0 : Assume that there is no association between the two variables.

Vs

H_1 : Assume that there is an association between the two variables.

Decision Criteria: Reject H_0 if the Chi-Square statistic is greater than the critical value.

i.e. $\chi^2_{(cal)} > \chi^2_{(Tab)}$ Otherwise accept H_0 .

2.2. PROPORTION TEST

The proportion test (or *z-test for proportions*) is used to compare proportions between groups or against a hypothesized value. It determines whether observed differences in proportions are statistically significant.

When to Use a Proportion Test?

- Compare a sample proportion to a known value (e.g., "Does 60% of our sample prefer Product A?").
- Compare two independent proportions (e.g., "Is the click-through rate different between Group A and Group B?").

What Does It Test?

- Compares sample proportions to:
 - A hypothesized population value (one-sample test)
 - Another sample's proportion (two-sample test)

Key Terms

- Proportion (p): Fraction of "successes" (e.g., $0.6 = 60\%$).
- Null Hypothesis (H_0): Assumes no difference in proportions.
- Alternative Hypothesis (H_1): Claims a significant difference exists.

Types of Proportion Tests

1. One-Sample Z-Test for Proportion

Purpose: Test if a sample proportion ($\hat{p}$) matches a theoretical value (p_0).

Hypotheses:

- $H_0: p = p_0$
- $H_1: p \neq p_0$

Formula:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Where:

$$\hat{p} = \frac{\text{Number of successes}}{n}$$

p_0 = Hypothesized proportion

n = Sample size

Example:

- Claim: "75% of users prefer Dark Mode."
- Sample: 80/100 users prefer it ($p=0.8$).
- Test if 0.8 differs significantly from 0.75.

2. Two-Sample Z-Test for Proportions

Purpose: Compare proportions between two independent groups.

Hypotheses:

- $H_0: p_1=p_2$
- $H_1: p_1 \neq p_2$

Formula:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Where:

- p_1, p_2 = Sample proportions
- $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$ pooled proportions
- n_1, n_2 = Sample sizes

Example:

- Group A: 50/100 users convert ($p_1=0.5$).
- Group B: 70/100 users convert ($p_2=0.7$).
- Test if the difference is significant.

3. Assumptions

1. Binary Data: Outcomes must be yes/no, success/failure.
2. Random Sampling: Observations are independent.
3. Sample Size:
 - Rule of Thumb: $np \geq 10$ and $n(1-p) \geq 10$.
 - If violated, use Fisher's Exact Test (for small samples).

5. Interpretation Guide

- $p\text{-value} < 0.05$: Reject H_0 (significant difference).
- Confidence Intervals: Check if the interval includes p_0 (one-sample) or 0 (two-sample).

6. Common Pitfalls

Small Samples: Use Fisher's Exact Test if $np < 10$ or $n(1-p) < 10$.

Non-Binary Data: Use Chi-Square or logistic regression.

Ignoring Effect Size: A statistically significant result may not be *practically* meaningful.

2.3. PEARSON CORRELATION

The Pearson Correlation Coefficient (Pearson's r) is a statistical measure that quantifies the linear relationship between two continuous variables. It assesses how well the relationship between the variables can be described by a straight line.

1. Conceptual Foundations

What Does Pearson's r Represent?

- Direction of Association:
 - Positive Correlation ($r > 0$): As one variable increases, the other tends to increase.
 - Negative Correlation ($r < 0$): As one variable increases, the other tends to decrease.
 - Zero Correlation ($r \approx 0$): No linear relationship exists.
- Strength of Association:
 - $0 \leq |r| < 0.3$: Weak correlation.
 - $0.3 \leq |r| < 0.7$: Moderate correlation.
 - $0.7 \leq |r| \leq 1$: Strong correlation.

Mathematical Definition

Pearson's r is the covariance of the two variables divided by the product of their standard deviations:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Where:

- $\text{Cov}(X, Y)$ = Joint variability of X and Y
- σ_X, σ_Y = Standard deviations of X and Y

This standardization ensures that r is scale-invariant (always between -1 and 1).

2. Assumptions

For Pearson's r to be valid, the following conditions must hold:

1. Continuous Variables

- Both variables should be measured on an interval or ratio scale.
- Not suitable for ordinal or categorical data.

2. Linearity

- The relationship between X and Y should be linear (not curved).
- Checked via a scatter plot (if points follow a straight line, linearity holds).

3. Normality

- Both variables should be approximately normally distributed.
- Can be checked using:
 - Histograms
 - Shapiro-Wilk test
 - Q-Q plots

4. Homoscedasticity (Equal Variance)

- The spread of residuals should be constant across all predicted values.
- Violations lead to unreliable correlation estimates.

5. No Significant Outliers

- Extreme values can distort the correlation coefficient.
- Example: A single outlier can turn a weak correlation into a strong one.

3. Hypothesis Testing with Pearson's r^*

Null Hypothesis (H_0)

- H_0 : There is no linear correlation between X and Y ($r^* = 0$).

Alternative Hypothesis (H_1)

- H_1 : There is a linear correlation ($r^* \neq 0$).

Test Statistic and p-value

- The p-value determines whether the correlation is statistically significant.
 - $p < 0.05$: Reject H_0 (significant correlation).
 - $p \geq 0.05$: Fail to reject H_0 (no significant evidence of correlation).

Degrees of Freedom (df)

- For Pearson's r^* , $df = n - 2$ (where n^* = sample size).

4. Interpretation of Results

Case 1: Strong Positive Correlation ($r \approx 0.9$)

- Meaning: A near-perfect linear increase in Y as X increases.
- Example: Study hours vs. exam scores.

Case 2: Moderate Negative Correlation ($r \approx -0.5$)

- Meaning: As X increases, Y tends to decrease, but not perfectly.
- Example: Exercise frequency vs. body fat percentage.

Case 3: Near-Zero Correlation ($r \approx 0.1$)

- Meaning: No meaningful linear relationship.
- Example: Shoe size vs. IQ.

5. Limitations and Misinterpretations

Limitations

1. Only Measures Linear Relationships

- Fails to detect non-linear patterns (e.g., quadratic, exponential).
- Solution: Use Spearman's rank correlation for monotonic (but not necessarily linear) relationships.

2. Sensitive to Outliers

- A single extreme value can inflate or deflate r .
- Solution: Check scatter plots for outliers before computing r .

3. Does Not Imply Causation

- Correlation $\neq$ Causation.
- Example: Ice cream sales and drowning incidents may correlate (both increase in summer), but one does not cause the other.

7. Practical Applications

- Psychology: Relationship between stress and sleep quality.
- Economics: GDP growth vs. unemployment rate.
- Medicine: Drug dosage vs. patient recovery time.
- Marketing: Ad spending vs. sales revenue.

2.4. MANN-WHITNEY U TEST:

The Mann-Whitney U test (also called the Wilcoxon rank-sum test) is a non-parametric statistical test used to compare differences between two independent groups when the dependent variable is either ordinal or continuous but not normally distributed. It serves as an alternative to the independent samples t-test when parametric assumptions are violated.

When to Use the Mann-Whitney U Test?

- Non-normal data: When the data violates the normality assumption.
- Ordinal data: When the dependent variable is ranked (e.g., survey Likert scales).
- Small sample sizes: Robust for small datasets where the t-test may fail.
- Unequal variances: Does not assume homogeneity of variance (unlike the t-test).

Hypotheses

- Null Hypothesis (H_0): The distributions of both groups are equal.
- Alternative Hypothesis (H_1): The distributions are not equal (two-tailed) or one group is stochastically greater (one-tailed).

2. Assumptions

For valid results, the test requires:

1. Independent Samples: The two groups must be unrelated (e.g., Group A vs. Group B).
2. Ordinal or Continuous Data: The dependent variable should be at least ordinal.
3. Same Shape Distributions: The distributions should have similar shapes (but can differ in location/median).

Note: Unlike the t-test, it does not require:

- Normality
- Homogeneity of variance
- Interval-scale data

3. How the Test Works

Step-by-Step Procedure

1. Combine and Rank Data:
 - Pool all observations from both groups.
 - Assign ranks from 1 (smallest) to N (largest).
 - Handle ties by assigning average ranks.

2. Calculate U Statistic:

- For each group, compute:

$$U_1 = R_1 - \frac{n_1(n_1 + 1)}{2}, \quad U_2 = R_2 - \frac{n_2(n_2 + 1)}{2}$$

Where:

- R_1, R_2 = Sum of ranks for Group 1 and Group 2.
- n_1, n_2 = Sample sizes of the two groups.
- The test statistic U is the smaller of U_1 or U_2 .

3. Compare U to Critical Values:

- For small samples ($n \leq 20$), use Mann-Whitney U tables.
- For larger samples ($n > 20$), U approximates a normal distribution

$$z = \frac{U - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}}$$

4. Make a Decision:

- Reject H_0 if $U \leq$ critical value or if $p < \alpha$ (typically 0.05).

4. Interpretation of Results

- Significant U: Suggests a difference in medians (or distributions).
- Effect Size: Common measures include:
 - Cliff's Delta: $(d = \frac{2U}{n_1 n_2} - 1)$.
 - Rank-Biserial Correlation (strength of association).

5. Advantages Over the t-Test

Robust to outliers (uses ranks, not raw data).

Works for ordinal data (t-test requires interval/ratio).

No normality assumption required.

6. Limitations

Less powerful than the t-test when data is normal.

Only compares medians if distributions have the same shape.

Not suitable for paired/repeated measures (use Wilcoxon signed-rank instead).

2.5. MULTINOMIAL LOGISTIC REGRESSION

Definition

Multinomial Logistic Regression is a type of generalized linear model used when the dependent variable is categorical and has more than two nominal (unordered) categories. It is an extension of binary logistic regression for multiclass classification problems

Multinomial Logistic Regression

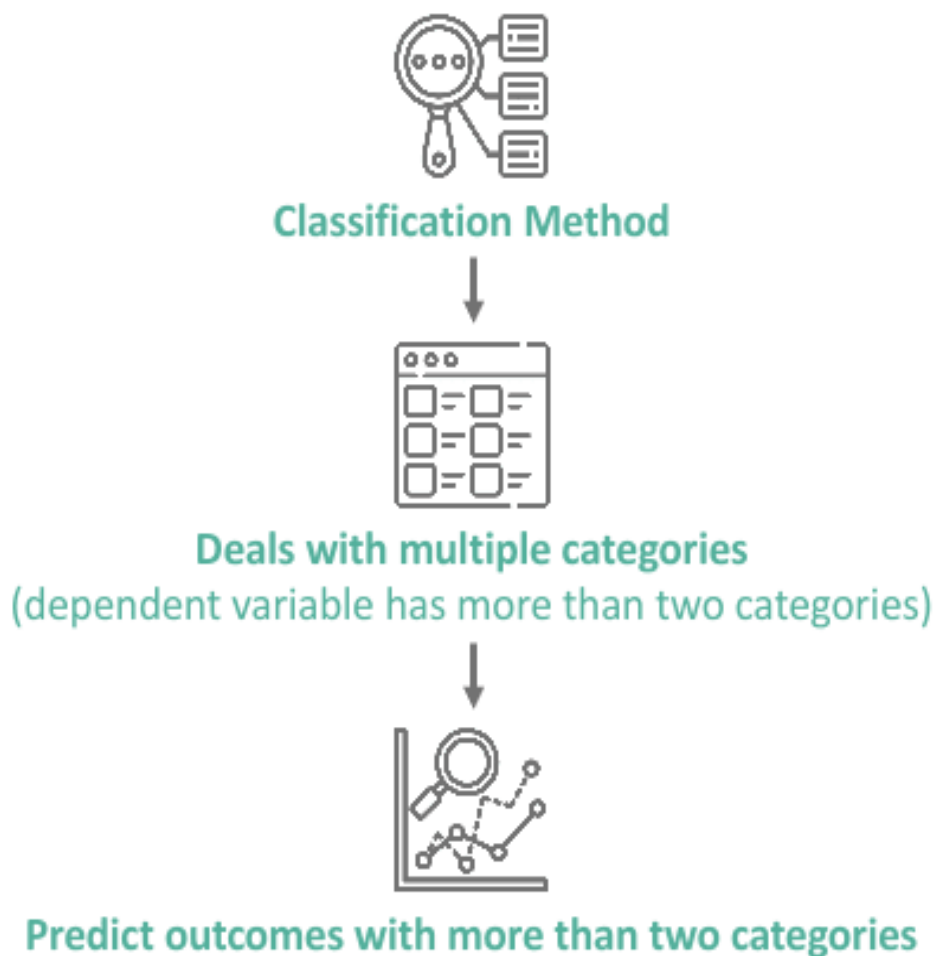


Chart No.2.5

Purpose

1. Model the relationship between a categorical outcome variable with $k > 2$ categories, and
2. One or more independent variables (continuous or categorical),
3. By estimating the probability of the different possible outcomes.

When to Use It

1. The dependent variable has three or more unordered categories.
2. Independent variables can be numerical, categorical, or both.
3. The dependent variable is nominal (not ordinal).

Model Structure

Let the dependent variable Y have k categories (e.g., $Y \in \{1, 2, \dots, k\}$). One category is set as the reference category (usually the last one, k).

We model the log odds of each category j relative to the reference category:

$$\log(P(Y = j)/P(Y = k)) = \beta_{0j} + \beta_{1j}X_1 + \beta_{2j}X_2 + \dots + \beta_{pj}X_p \quad \text{for } j = 1, 2, \dots, k-1$$

Interpretation of Coefficients

1. Each coefficient β_{ij} represents the log odds change of choosing category j over the reference category k for a unit change in X_i .
2. Coefficients are relative to the reference category.
3. Positive coefficients: Higher odds of being in category j vs. reference.
4. Negative coefficients: Lower odds of being in category j vs. reference.

Assumptions of MLR

1. Independence of Irrelevant Alternatives (IIA): The odds between any two categories are independent of other alternatives.
2. No multicollinearity: Independent variables should not be highly correlated.
3. Large sample size: MLR requires a sufficient number of observations per category.
4. Linearity in the logit: The relationship between independent variables and the log odds is linear.

Model Evaluation Metrics

1. Log-likelihood: Measures model fit.
2. Pseudo R^2 (McFadden's R^2): An analog to R^2 in linear regression.
3. Likelihood Ratio Test: Compares full model vs. null model.
4. Confusion Matrix: For classification accuracy.
5. AIC/BIC: Used for model comparison.

Limitations

- 1 .Violation of IIA can lead to biased estimates.
2. Difficult to interpret with many categories or predictors.
- 3 Sensitive to sample size.
4. Results can be unstable if one category has very few cases.

Applications

1. Market research (predict customer choices)
2. Healthcare (predict disease types)
3. Political science (predict party affiliation)
4. Education (predict course selection)

2.6. KRUSKAL-WALLIS TEST

Kruskal-Wallis Test refers to a method of matching the medians of more than two groups to ascertain whether the samples have the same group source as an origin or not. It is a key tool for comparing three or more groups based on a dependent variable by measuring at a categorical level.

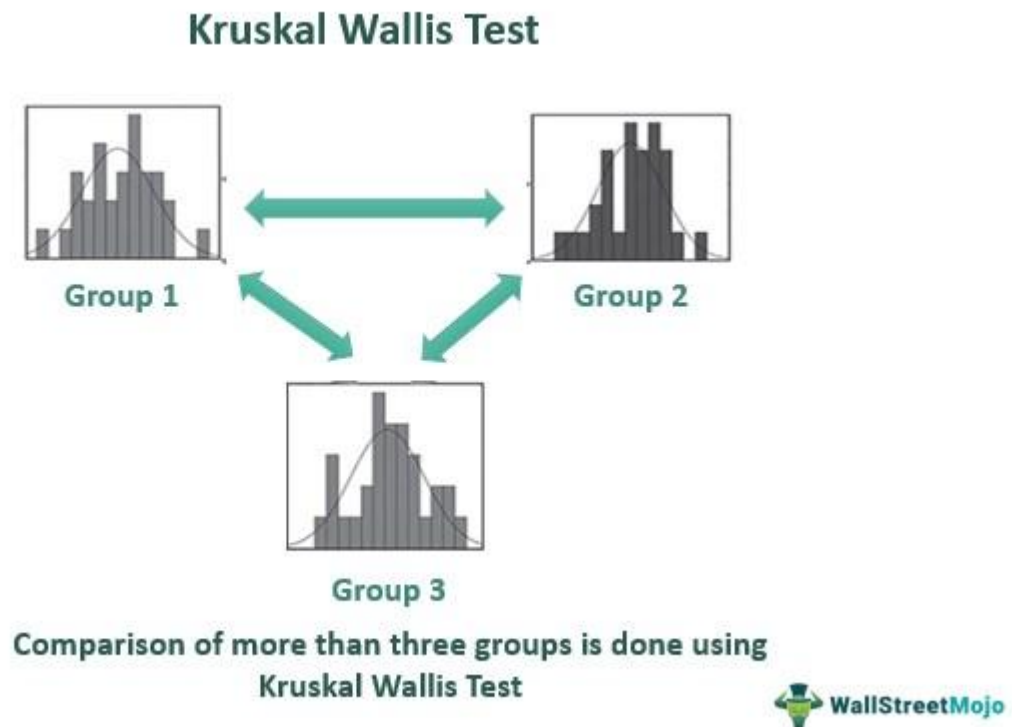


Chart No.2.6

It applies to the distribution of less or non-parametric populations for contrasting more than two distinct and equal-sized data samples. This test examines the null hypothesis, which claims that 'k' samples from the same population had identical median values. It indicates the dominance of one sample of the variable over the other sample stochastically.

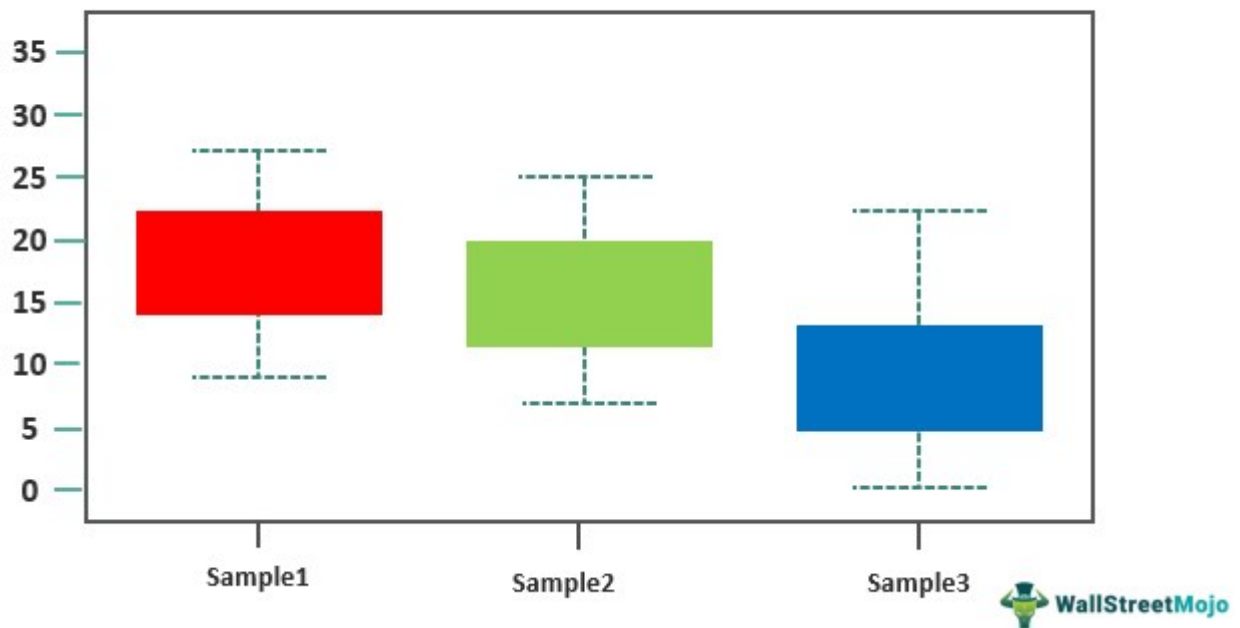
Key Takeaways

- Kruskal-Wallis Test is a method to determine the dominance of a variable of one sample over a variable of another sample when the number of groups is more than three by comparison of their median in a parametric manner.
- It has certain determinants, like more than three conditions and the absence of normally distributed data to apply the test.
- It is used only when more than three groups have independent variables and more than three categories of data.
- The value of p is always more than 0.05, indicating little power of the test.

Kruskal-Wallis Test:

Kruskal-Wallis H test is a type of one-way ANOVA test based on ranks that uses statistical methods to compare the medians of more than two groups. So, one can apply it to all distributions containing variables that are either ordinal or dependent continuous levels. However, although it can establish which variables dominate within a group, it fails to answer the reason for dominance.

Kruskal Wallis Test - Overview



The test in consideration is applicable only when the following conditions are place:

- One has more than three conditions for comparison.
- Every condition is regulated by a distinct participant group, meaning that if one has the design of independence measures, having more than three conditions.
- The data for applying the test is not distributed normally.
- The data being measured have markedly different variances related to different conditions.

After the above conditions are met, the test is performed to check whether the outcome of one group under the test affects the outcome of the other sample under consideration.

Advertisement

The test assesses various data sets for understanding the interchange of data samples among numerous data sets. Then the null hypothesis is utilized to know whether the medians are equal, as opposed to the alternative hypothesis, which suggests the difference in at least one of the samples exists. Machine learning also takes the help of the test to ascertain the difference between two or more groups without telling the reason for the existing difference between them.

Factors

Let us understand the concept by learning how test anxiety influences test results. Three possible values for the independent variable "test anxiety" exist –

1. No anxiety,
2. Low-medium anxiety, and
3. High anxiety.

The dependent variable is the exam score, scaled from 0 to 100%.

Another example is learning how socioeconomic status influences perceptions of increasing sales tax. Socioeconomic status has three tiers:

1. Working class,
2. Middle class, and
3. Wealthy,

These are the independent variables. The dependent variable is measured on a 5-point Likert scale, from strongly agree to strongly disagree.

Assumptions

The researcher's variables ought to include the following Kruskal-Wallis test assumptions:

- Two or more tiers and one independent variable (independent groups). The test is more frequently administered when statisticians have three or more levels. Consider utilizing the Mann-Whitney U Test for two levels instead.
- Dependent variables with an interval, ratio, or ordinal scale.
- Statisticians' observations ought to be impartial. Put another way, it shouldn't have any connections between the individuals who make up each group or between groupings.
- The distributions of shapes for all groups ought to be uniform. Most testing tools, including SPSS and Minitab, will check for this condition.

Formula

Let us determine the Kruskal-Wallis test formula for comparing medians of more than two groups. Below is the formula for the test:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1)$$

Where,

- K = number of groups used for comparison
- N = total size of the sample

- n_i = i th group's sample size
- R_i = total of the ranks related to the i -th group

The above formula is valid only for a group of samples:

- Having a minimum of five elements.
- Group having no estimated population barriers.
- The distribution of the population has no assumption.
- Sample groups are independent.
- Random selection of data in every group.
- Having a very minimal ordinal.

Moreover, certain websites and software provide Kruskal-Wallis test calculators for solving the formula.

2.7.Political Awareness in Research Project

Introduction to Political Awareness

Political awareness refers to an individual's understanding of the political environment, including knowledge about political parties, leaders, elections, government structures, and current political events. It reflects a person's engagement with political matters and their ability to make informed decisions as a citizen. Political awareness is a critical aspect of a healthy democracy, as it influences voter behavior, civic participation, and overall political stability.

Importance of Political Awareness

Political awareness is essential because it:

- Encourages informed voting and political participation.
- Reduces the impact of misinformation and propaganda.
- Promotes accountability and transparency in governance.
- Strengthens democratic processes by empowering citizens.

Measuring Political Awareness

In this project, political awareness is measured through a structured questionnaire containing a series of multiple-choice and identification questions. Each question is designed to assess the respondent's knowledge and understanding of political processes, leaders, symbols, and current political scenarios.

Awareness Scoring Method

Each correct answer is awarded 1 mark. The total awareness score is the sum of all correct responses. The final awareness score can be expressed as a percentage using the formula:

$$\text{Awareness Score (\%)} = (\text{Total Correct Answers} / \text{Total Questions}) \times 100$$

Example Calculation

For a questionnaire with 19 questions:

- If a respondent answers 9 questions correctly, their awareness score would be:

$$\text{Awareness Score} = (9 / 19) \times 100 = 47\%$$

Questions Used to Measure Political Awareness

1. Are you interested in politics?
2. How often do you follow political news?
3. Where do you get your political information from?
4. Are you affiliated with any political party?
5. Would you consider a career in politics?
6. Did you vote in the recent election?
7. What kind of election happened recently?
8. How often are Lok Sabha elections held?
9. Identify the political party of the following leaders:
10. Match the following party symbols:

11. Do you believe elections are transparent and fair?
12. Do you think party manifestos are truthful?
13. Do you think that politicians changing parties is acceptable?
14. Are you satisfied with the current political situation in your state?
15. Should young people consider politics as a career?
16. Are you interested in politics?
17. How often do you follow political news?
18. Where do you get your political information from?
19. Are you affiliated with any political party?

2.9 ENTRANCE POLLS

Entrance polls are a critical part of modern political analysis, conducted immediately after voters exit the polling stations. They serve as a powerful tool for predicting the outcomes of elections before the official results are declared. Entrance polls capture real-time voter preferences and behavior, offering insights into voting patterns, key issues influencing decisions, and the potential direction of the political landscape. They provide an early indication of the likely winners and the overall mood of the electorate.

Purpose of Entrance polls

The primary objectives of entrance polls include:

- Estimating the likely outcome of an election.
- Understanding voter preferences and motivations.
- Identifying the most influential factors in voter decision-making.
- Assessing the performance of political parties and alliances.
- Analyzing regional variations in voting behavior.

Methodology for Entrance polls

Entrance polls are typically conducted using structured questionnaires that capture the following key data points:

- Whether the voter participated in the election.
- The party the voter believes will win in their constituency.
- The factors considered while selecting a candidate.
- The preferred candidate for the next Chief Minister.
- The alliance most likely to come to power.
- Voter perceptions about the fairness and transparency of the election.

The responses collected are then aggregated, analyzed, and interpreted to project the probable outcomes of the election.

Entrance poll Questions Used in This Project

1. Did you vote in the recent elections?
2. Which party do you think will win in your constituency?
3. What factors do you consider when voting for a candidate?
4. Who do you think should be the next Chief Minister?
5. Which alliance do you think will come to power?
6. Do you think the recent elections were fair and transparent?

Conclusion

Entrance polls, while not always perfectly accurate, provide valuable early insights into the political pulse of the electorate. They contribute to the democratic process by enhancing transparency and public awareness about election outcomes. However, their accuracy can be influenced by factors like sample size, regional biases, and voter reluctance to disclose true preferences.

4. Analysis and results

3.1 CHI-SQUARE

3.1.1 Association between gender and political awareness.

The Chi-square test of independence is a statistical method used to check if there is any association or relationship between two categorical variables. It helps us understand whether changes in one variable (like gender) are related to changes in another variable (like political awareness).

This study measured political awareness using a set of 19 questions. These questions asked about political knowledge, interest in politics, and voting. etc

Each participant got a total score based on their answers. Then, they were grouped as:

Aware: Score of 9 or more out of 19

Not Aware: Score less than 9

This method is based on the study by Ahmed et al. (2015), who used a similar way to measure political awareness among university students in Pakistan.

Reference:

Ahmed, Z., Javaid, M. A., Muzaffar, M., Fatima, N., & Hussain, T. (2015). Comparing the level of political awareness among the students of natural and social sciences: A case study of public sector universities in Pakistan. *Pakistan Journal of Life and Social Sciences*, 13(2), 64–67.

Hypothesis:

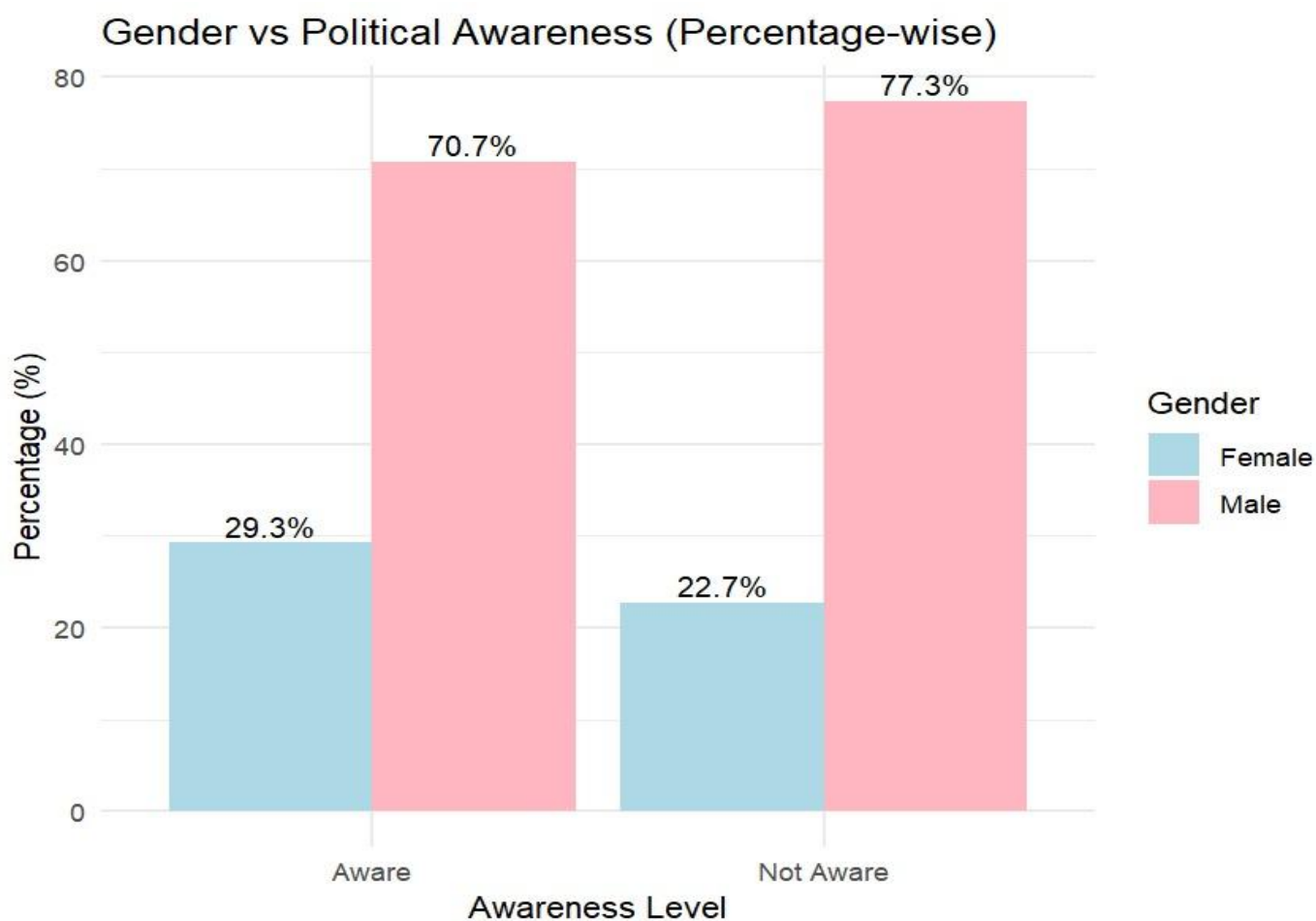
Null Hypothesis (H_0):

There is no association between gender and political awareness.

Alternative Hypothesis (H_1):

There is an association between gender and political awareness.

Graphical representation :



Graph 3.1

The bar graph No. 3.1 shows the relationship between Gender and Political Awareness levels among people. The graph shows that males make up a larger share of both aware and not aware groups, indicating their overall dominance in the sample.

R. code:

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
+ - Go to file/function Addins
Untitled2 x Untitled23* x Untitled24* x Untitled25* x Untitled26* x Untitled27* x Untitled28* x Un>>
Source on Save Run Source
1
2 data <- read_excel("data 2.xlsx")
3 data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
4 names(data)
5 table_data <- table(data$Gender, data$Awareness)
6 print(table_data)
7 chisq.test(table_data)
8
9 data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
10
11 library(ggplot2)
12
13 # Convert to data frame for ggplot
14 plot_data <- as.data.frame(table(data$Gender, data$Awareness))
15 colnames(plot_data) <- c("Gender", "Awareness", "Count")
16
17 # Change gender to labels
18 plot_data$Gender <- factor(plot_data$Gender, levels = c("0", "1"),
19                           labels = c("Female", "Male"))
20
21 # Plot
22 ggplot(plot_data, aes(x = Awareness, y = Count, fill = Gender)) +
23   geom_bar(stat = "identity", position = "dodge") +
24   labs(title = "Gender vs Political Awareness",
25        x = "Awareness Level",
26        y = "Count") +
27   scale_fill_manual(values = c("lightblue", "lightpink")) +
28   theme_minimal()
29
30
```

Output:

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
+ - Go to file/function Addins
Source
Console Background Jobs x
R 4.3.1 ~ /
> getwd()
[1] "C:/Users/ar/OneDrive/Documents"
>
> library(readxl)
> data <- read_excel("data 2.xlsx")
> data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
> names(data)
[1] "Gender" "Score" "Awareness"
>
>
> table_data <- table(data$Gender, data$Awareness)
> print(table_data)
      Aware Not Aware
0      134         32
1      324        109
>
> chisq.test(table_data)

Pearson's Chi-squared test with Yates' continuity correction

data:  table_data
X-squared = 2.0017, df = 1, p-value = 0.1571
>
>
> chisq.test(table_data)

Pearson's Chi-squared test with Yates' continuity correction

data:  table_data
X-squared = 2.0017, df = 1, p-value = 0.1571
>
>
> data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
>
```

$\chi^2 = 2.0017$.

df = 1

p-value = 0.1571.

If the p-value is less than 0.05, the result is considered significant, indicating a real difference. In this case, the p-value is 0.1571, which is greater than 0.05; therefore, the null hypothesis (H_0) is accepted.

3.1.2. Association Between Locality And Political awareness

Type of Test Used in This Study

In this study, the Chi-Square Test of Independence was used to examine the relationship between:

Data Description:

Locality (0 = Rural, 1 = Urban)

Political Awareness ("Aware", "Not Aware")

Sample Size: 599 respondents

Locality Distribution:

Rural: 180 (30.1%) Urban: 419 (69.9%)

Political Awareness Classification:

Aware: Score ≥ 9 (out of 19)

Not Aware: Score < 9 (out of 19)

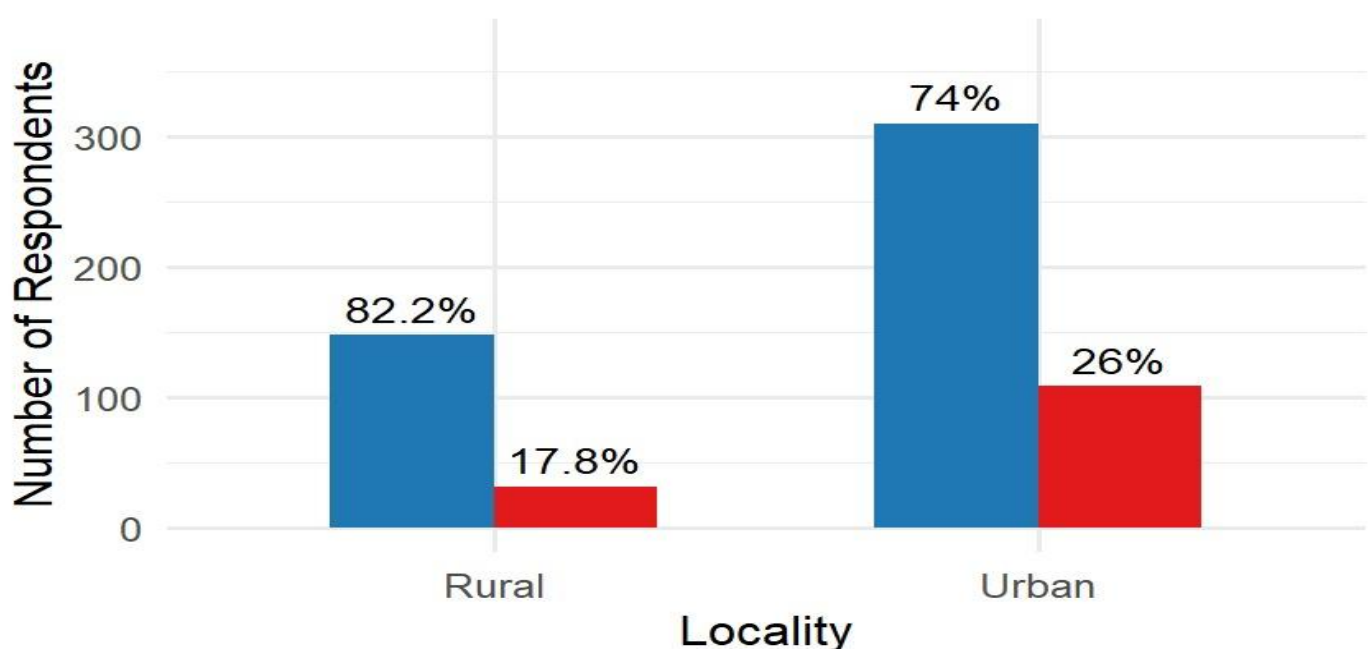
- Refer to Section 3.1.1 for detailed classification criteria.

Data Source: Primary data collected through a survey.

Graphical representation :

Political Awareness by Locality

Political Awareness ■ Aware ■ Not Aware



Graph 3.2

Graph no 3.2 shows political awareness among respondents in rural and urban areas. In the rural area, 82.2% of respondents are aware, while 17.8% are not. In the urban area, 74% of respondents are aware, and 26% are not aware. This indicates that political awareness is higher in rural areas compared to urban areas.

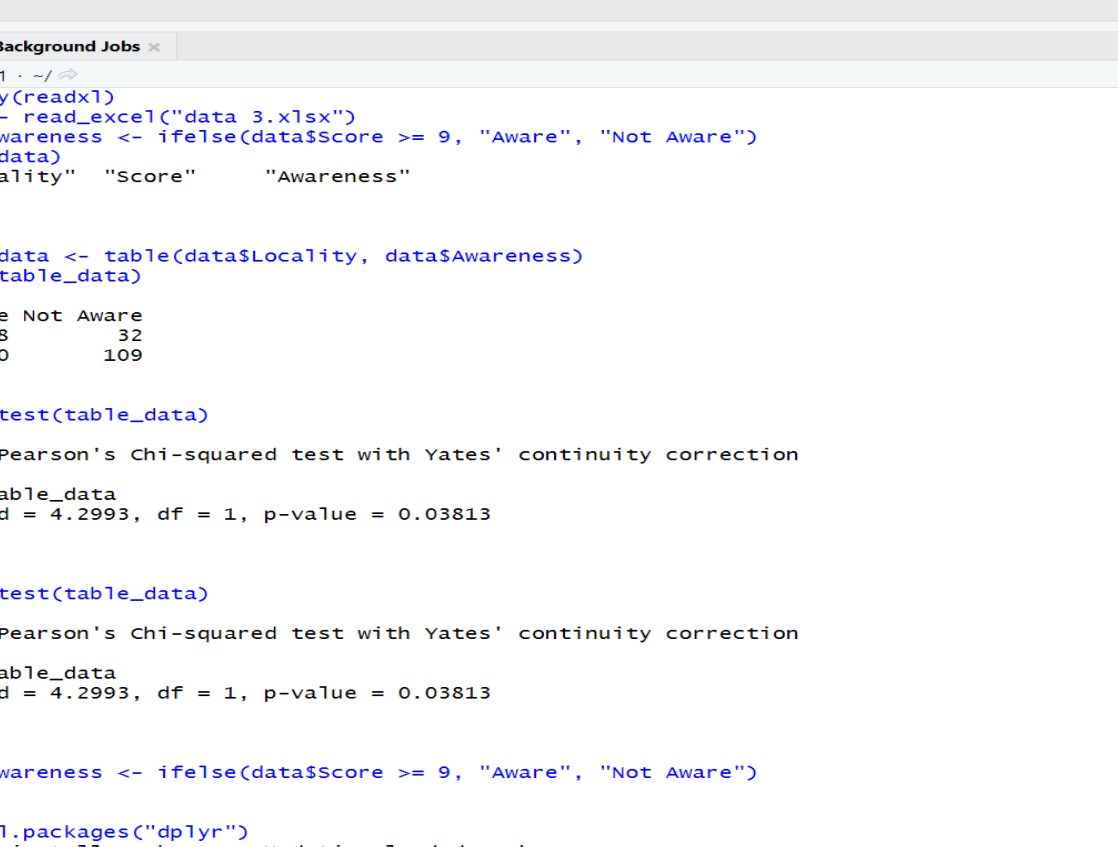
Hypothesis

1. Null Hypothesis (H_0): There is no association between locality and political awareness.
2. Alternative Hypothesis (H_1): There is an association between locality and political awareness.

R. code :

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
+ - + - + - Go to file/function Addins
Untitled21* x Untitled22* x Untitled23* x Untitled24* x
Source on Save Run Source
1 library(readxl)
2 data <- read_excel("data 3.xlsx")
3 data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
4 names(data)
5 table_data <- table(data$Locality, data$Awareness)
6 print(table_data)
7 chisq.test(table_data)
8 data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
9 install.packages("dplyr")
10 library(dplyr)
11 # Load libraries
12 library(readxl)
13 library(ggplot2)
14 library(dplyr)
15
16 # Read data
17 data <- read_excel("data 3.xlsx")
18 # Check column names
19 names(data)
20 # Create Awareness column
21 data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
22 # Convert Locality to factor for plotting
23 data$Locality <- factor(data$Locality, levels = c(0, 1), labels = c("Rural", "Urban"))
24 # Summarize data
25 summary_data <- data %>%
26   group_by(Locality, Awareness) %>%
27   summarise(Count = n(), .groups = "drop")
28 # View summary to verify
29 print(summary_data)
30 # Grouped bar chart
31 ggplot(summary_data, aes(x = Locality, y = Count, fill = Awareness)) +
32   geom_bar(stat = "identity", position = "dodge", width = 0.6) +
33   scale_fill_manual(values = c("Aware" = "#1f78b4", "Not Aware" = "#e31a1c")) +
34   labs(
35     title = "Political Awareness by Locality",
36     x = "Locality",
37     y = "Number of Respondents",
38     fill = "Political Awareness"
39   ) +
40   theme_minimal(base_size = 14) +
41   theme(
42     plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
43     legend.position = "top"
44   )
45
```

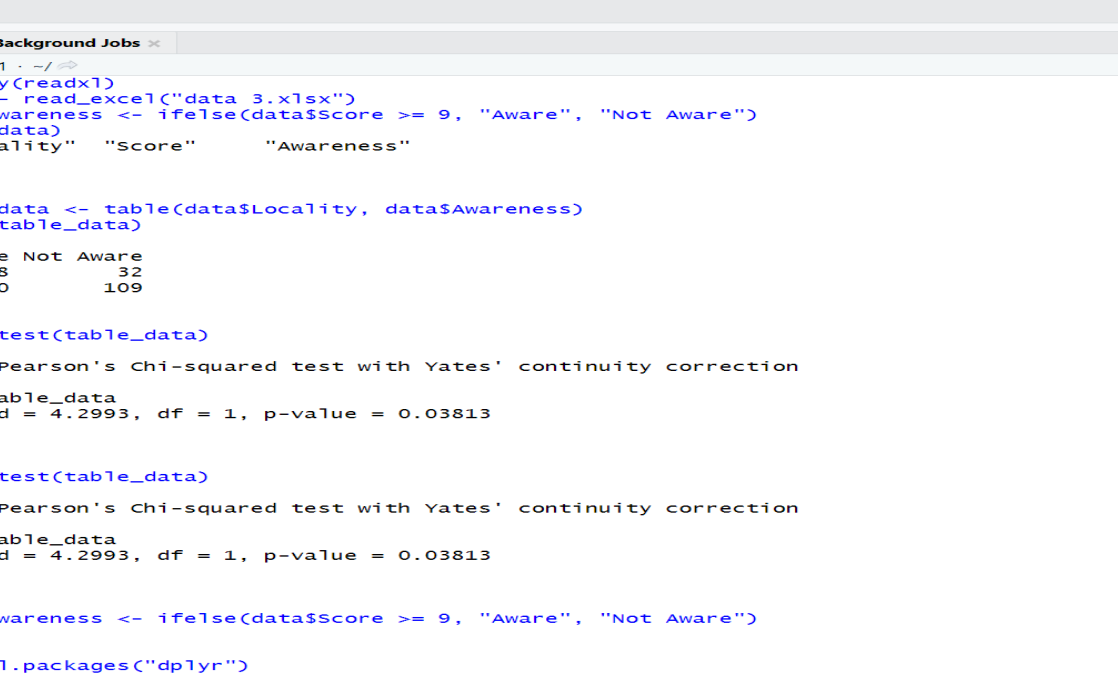
Output



The screenshot shows the RStudio interface with the following components:

- Menu Bar:** File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, Help.
- Toolbar:** Includes icons for saving, running, and navigating, along with a "Go to file/function" search bar and an "Addins" dropdown.
- Source Panel:** Displays the R script being executed. The script includes the following code:


```
> library(readxl)
> data <- read_excel("data 3.xlsx")
> data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
> names(data)
[1] "Locality" "Score"      "Awareness"
>
>
> table_data <- table(data$Locality, data$Awareness)
> print(table_data)
      Aware Not Aware
0      148        32
1      310       109
>
> chisq.test(table_data)
      Pearson's Chi-squared test with Yates' continuity correction
data:  table_data
X-squared = 4.2993, df = 1, p-value = 0.03813
>
>
> chisq.test(table_data)
      Pearson's Chi-squared test with Yates' continuity correction
data:  table_data
X-squared = 4.2993, df = 1, p-value = 0.03813
>
>
> data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
>
>
> install.packages("dplyr")
Error in install.packages : Updating loaded packages
> library(dplyr)
> # Load libraries
> library(readxl)
> library(ggplot2)
> library(dplyr)
```
- Console Panel:** Shows the output of the script, including the table structure and the results of the chi-squared test.



The screenshot shows the RStudio interface with the console window active. The code in the console is as follows:

```

> library(readxl)
> data <- read_excel("data 3.xlsx")
> data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
> names(data)
[1] "Locality" "Score"      "Awareness"
>
>
> table_data <- table(data$Locality, data$Awareness)
> print(table_data)
      Aware Not Aware
0      148         32
1      310        109
>
>
> chisq.test(table_data)

      Pearson's Chi-squared test with Yates' continuity correction

data:  table_data
X-squared = 4.2993, df = 1, p-value = 0.03813
>
>
> chisq.test(table_data)

      Pearson's Chi-squared test with Yates' continuity correction

data:  table_data
X-squared = 4.2993, df = 1, p-value = 0.03813
>
>
> data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
>
>
> install.packages("dplyr")
Error in install.packages : Updating loaded packages
> library(dplyr)
> # Load libraries
> library(readxl)
> library(ggplot2)
> library(dplyr)

```


1. Contingency Table:

	Aware	Not Aware
Rural (0)	148	32
Urban (1)	310	109

The table shows how people in rural and urban areas are aware of politics. In rural areas, 148 people are aware, and 32 are not. In urban areas, 310 people are aware, and 109 are not. This helps us see if people in rural and urban areas have different levels of political awareness.

$\chi^2 = 4.2993$), $df = 1$

p-value = 0.03813:

Since the p-value (0.03813) is less than 0.05, the null hypothesis is rejected.

Conclusion

There is a significant association between locality and political awareness.

This means that political awareness differs by locality — urban respondents are more politically aware compared to rural respondents in this sample.

3.1.3 . Association between income level and political awareness

The Chi-Square Test of Independence is a statistical test used to determine whether there is a significant association between two categorical variables. For the present study chi-square test is used to check the association between income level and political awareness

Variables:

- Income Level (Categorical, coded as 0 = Low, 1 = Medium Low, 2 = Medium High, 3 = High)
- Political Awareness (Categorized from Score as: Aware / Not Aware)

Data Description

This study uses a total sample of 599 respondents to analyze the relationship between income level and political awareness. The data for this analysis were collected through a structured survey, and the variables are defined as follows:

Income Level: Categorical variable with four levels:

1. Low (0): 370 respondents (61.8%)
2. Medium Low (1): 111 respondents (18.5%)
3. Medium High (2): 65 respondents (10.9%)
4. High (3): 53 respondents (8.8%)

Political Awareness: Categorical variable based on survey scores:

1. Aware: 458 respondents (76.5%)
2. Not Aware: 141 respondents (23.5%)

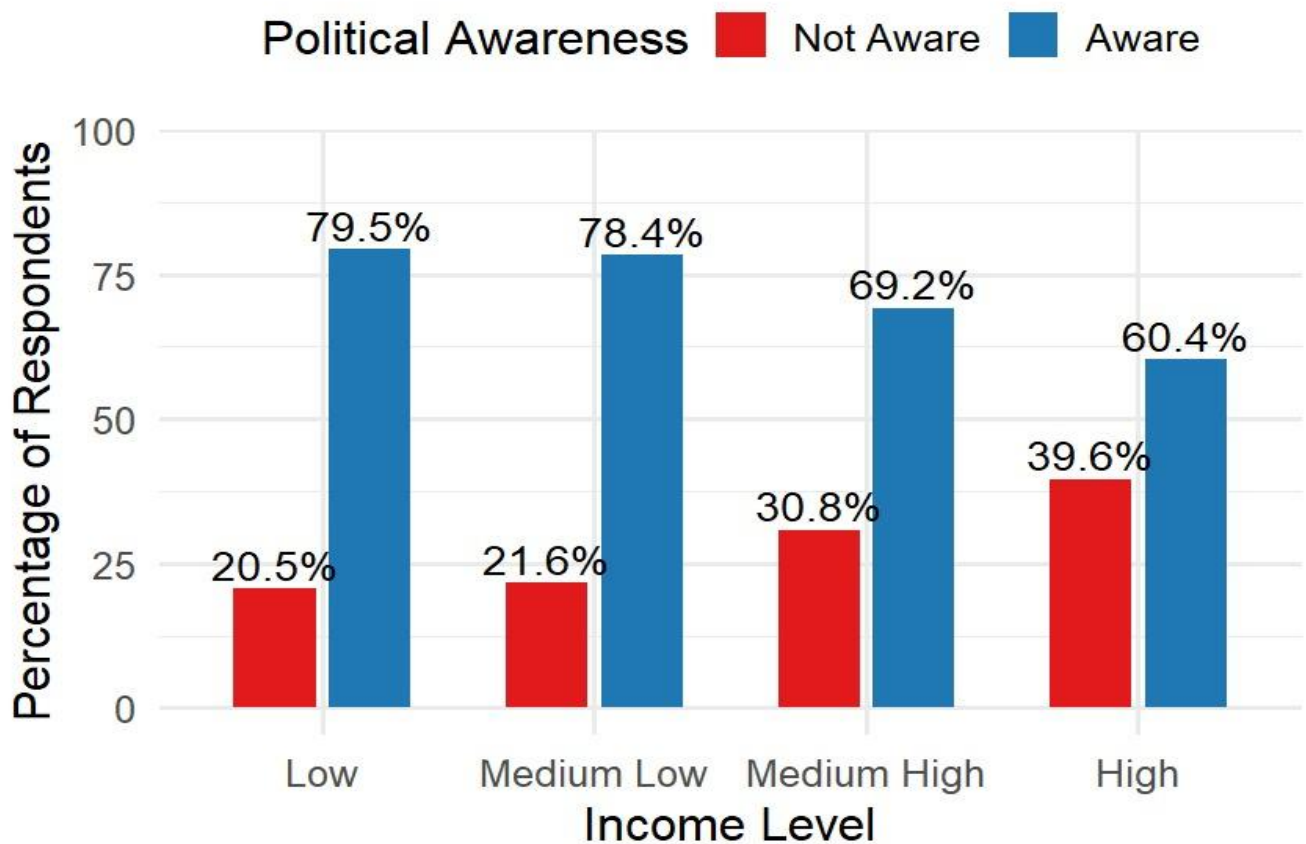
Source: The data was obtained from a structured survey conducted for the present study.

Hypotheses:

- Null Hypothesis (H_0): Income level and political awareness are independent (no association).
- Alternative Hypothesis (H_1): Income level and political awareness are dependent (there is an association).

Graphical representation :

Political Awareness by Income Level (Percent)



Graph 3.3

The bar graph No3.3 shows that political awareness tends to decrease as income level increases. Specifically:

1. Low Income: 79.5% are aware, 20.5% are not aware.
2. Medium Low Income: 78.4% are aware, 21.6% are not aware.
3. Medium High Income: 69.2% are aware, 30.8% are not aware.
4. High Income: 60.4% are aware, 39.6% are not aware.

This suggests that individuals with higher income levels have relatively lower political awareness compared to those with lower income.

Contingency Table:

Income Level	Aware	Not Aware	Total
0 (Low)	294	76	370
1 (Medium Low)	87	24	111
2 (Medium High)	45	20	65
3 (High)	32	21	53

R. code:

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Untitled21* x Untitled22* x Untitled23* x Untitled24* x Untitled25* x Untitled26* x Untitled27* x
Source on Save Run Source
1 # Load libraries
2 library(readxl)
3 library(ggplot2)
4 library(dplyr)
5
6 # Read data
7 data <- read_excel("data 4.xlsx")
8
9 # Check column names
10 names(data)
11 # Create Awareness column (same logic as before, based on score)
12 data$Awareness <- ifelse(data$Score >= 9, "Aware", "Not Aware")
13 # Convert Income and Awareness to factors for plotting
14 data$Income <- factor(data$Income, levels = c(0, 1, 2, 3), labels = c("Low", "Medium Low", "Medium High", "High"))
15 data$Awareness <- factor(data$Awareness, levels = c("Not Aware", "Aware"))
16
17 # Summarize data by Income and Awareness
18 summary_data <- data %>%
19   group_by(Income, Awareness) %>%
20   summarise(Count = n(), .groups = "drop")
21
22 # View summary to verify
23 print(summary_data)
24
25 # Grouped bar chart for Income and Awareness
26 ggplot(summary_data, aes(x = Income, y = Count, fill = Awareness)) +
27   geom_bar(stat = "identity", position = "dodge", width = 0.6) +
28   scale_fill_manual(values = c("Aware" = "#1f78b4", "Not Aware" = "#e31a1c")) +
29   labs(
30     title = "Political Awareness by Income Level",
31     x = "Income Level",
32     y = "Number of Respondents",
33     fill = "Political Awareness"
34   ) +
35   theme_minimal(base_size = 14) +
36   theme(
37     plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
38     legend.position = "top"
39   )
40
41 table_data <- table(data$Gender, data$Awareness)
42 print(table_data)
43 chisq.test(table_data)
44
45
46
```

Output:

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
+ [Go to file/function] [Addins]

Source

Console Background Jobs x
R 4.3.1 · ~/
+ summarise(Count = n(), .groups = "drop")
>
> # View summary to verify
> print(summary_data)
# A tibble: 8 × 3
  Income      Awareness Count
<fct>      <fct>    <int>
1 Low       Not Aware    76
2 Low       Aware       294
3 Medium Low Not Aware    24
4 Medium Low Aware      87
5 Medium High Not Aware    20
6 Medium High Aware     45
7 High      Not Aware    21
8 High      Aware     32
>
> # Grouped bar chart for Income and Awareness
> ggplot(summary_data, aes(x = Income, y = Count, fill = Awareness)) +
+   geom_bar(stat = "identity", position = "dodge", width = 0.6) +
+   scale_fill_manual(values = c("Aware" = "#1f78b4", "Not Aware" = "#e31a1c")) +
+   labs(
+     title = "Political Awareness by Income Level",
+     x = "Income Level",
+     y = "Number of Respondents",
+     fill = "Political Awareness"
+   ) +
+   theme_minimal(base_size = 14) +
+   theme(
+     plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
+     legend.position = "top"
+   )
> print(table_data)

  Aware Not Aware
0   294         76
1    87         24
2    45         20
3    32         21
> chisq.test(table_data)

Pearson's Chi-squared test

data:  table_data
X-squared = 11.58, df = 3, p-value = 0.008968
>
```

Interpretation:

The p-value $(0.0089) < 0.05$, so the null hypothesis is rejected.

Conclusion :

The Chi-Square Test of Independence ($\chi^2 = 11.58$, $df = 3$, $p = 0.009$) indicates a significant association between income level and political awareness ($p < 0.05$).

3.1.4. Chi-square test of independence between news time and political awareness score level

1. Objective of the Test

To determine whether there is a statistically significant association between individuals' news time and their categorized score levels, which were grouped into Low, Medium, and High.

Sample Size: The analysis is based on 598 respondents, categorized by their news viewing frequency and political awareness score levels.

Independent Variable (News time Frequency):

Daily: 326 respondents

Weekly: 94 respondents

Rarely: 178 respondents (includes those who view news "Sometimes" or "Never")

Dependent Variable (Score Group):

Low (0-6): 78 respondents

Medium (7-13): 262 respondents

High (14-19): 258 respondents

Source of Data: Collected through a structured survey.

Hypothesis

Null Hypothesis (H_0):

There is no association between news consumption frequency and score group; the variables are independent.

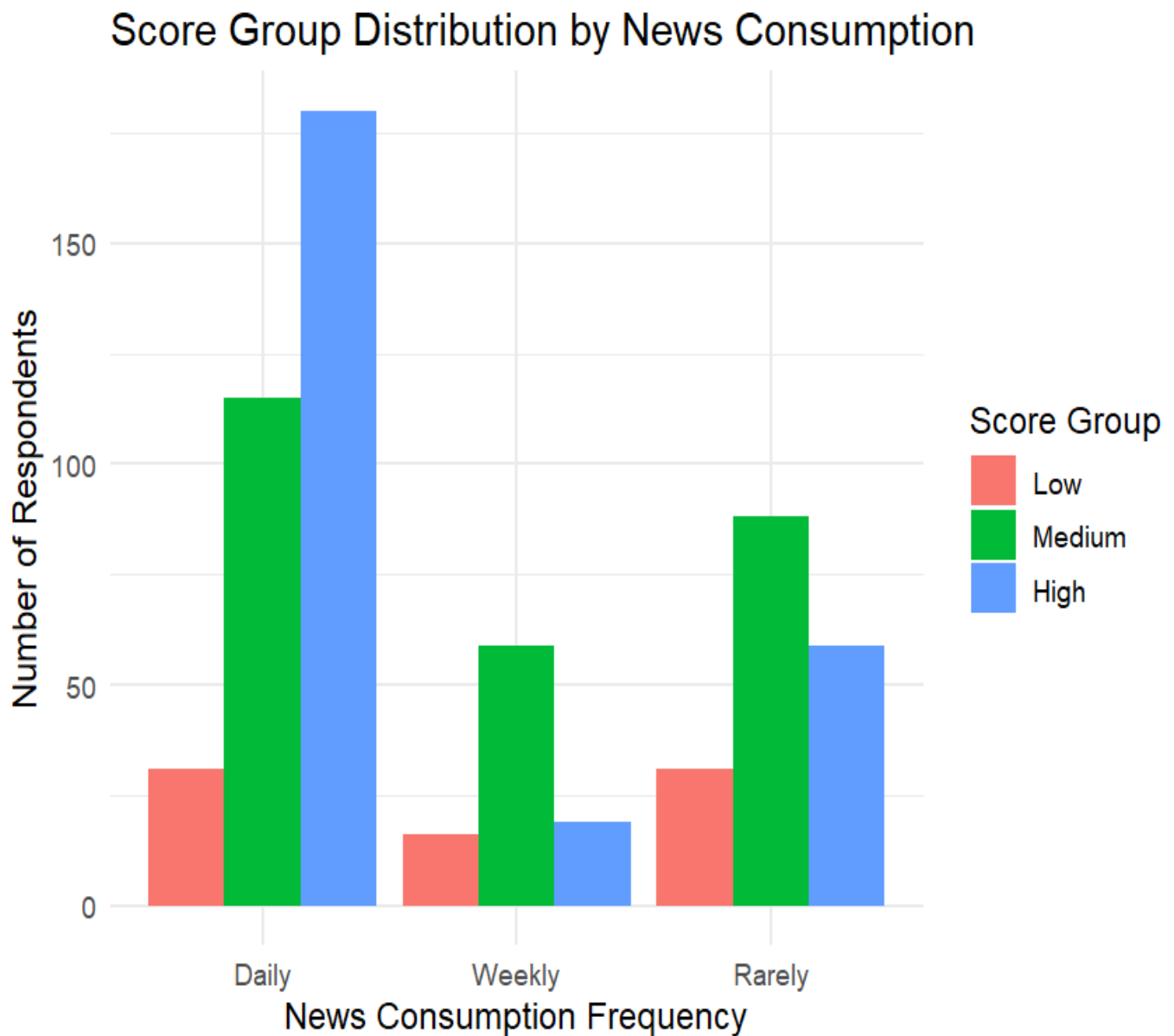
Alternative Hypothesis (H_1):

There is an association between news consumption frequency and score group; the variables are dependent.

Contingency Table :

News Consumption	Low	Medium	High	Total
Daily	31	115	180	326
Rarely	31	88	59	178
Weekly	16	59	19	94
Total	78	262	258	598

Graphical representation :



Graph 3.4

Conclusion:

People who read news daily are more likely to have higher scores. Those who read news weekly or rarely mostly fall in the medium score group. This suggests that frequent news consumption might be linked to higher scores .

R. code:

The screenshot shows the RStudio interface with the following components:

- Menu Bar:** File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, Help.
- Toolbar:** Includes icons for file operations (new, open, save, print), navigation (back, forward), and execution (run, source).
- Session Information:** Shows the current session is 'Source on Save'.
- File Explorer:** Displays several untitled files (Untitled33* through Untitled39*).
- Code Editor:** Contains the following R code:

```
1 library(readxl)
2
3
4 df = read_excel("data111.xlsx")
5 names(data)
6
7
8 data$ScoreGroup = cut(data$score,
9                       breaks = c(0, 6, 13, 19),
10                      labels = c("Low", "Medium", "High"),
11                      include.lowest = TRUE)
12
13 table_data = table(data$News, data$ScoreGroup)
14
15 chi_result = chisq.test(table_data)
16
17 print(table_data)
18 print(chi_result)
19
20
21 data$News_Mod = as.character(data$News)
22 data$News_Mod[data$News_Mod %in% c("Never", "Sometimes")] = "Rarely"
23 data$News_Mod = as.factor(data$News_Mod)
24
25 table_data_mod = table(data$News_Mod, data$ScoreGroup)
26 chi_result_mod = chisq.test(table_data_mod)
27
28 print(table_data_mod)
29 print(chi_result_mod)
30
31 library(ggplot2)
32 library(dplyr)
33
34 plot_data = data %>%
35   mutate(News_Mod = ifelse(News %in% c("Never", "Sometimes"), "Rarely", as.character(News)))
36   group_by(News_Mod, ScoreGroup) %>%
37   summarise(Count = n()) %>%
38   ungroup()
39
40 plot_data$News_Mod = factor(plot_data$News_Mod, levels = c("Daily", "Weekly", "Rarely"))
41 plot_data$ScoreGroup = factor(plot_data$ScoreGroup, levels = c("Low", "Medium", "High"))
42
```
- Search Bar:** Located above the code editor, with options for 'In selection', 'Match case', 'Whole word', 'Regex', and 'Wrap'.
- Console:** At the bottom, showing the current line (42:1) and the context '(Top Level)'.

Output:

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Source
Console Background Jobs x
R 4.3.1 ~ /
> library(readxl)
> df = read_excel("data111.xlsx")
> names(df)
[1] "Gender"      "Age"          "Education"    "Locality"    "Category"    "Income"
[7] "Score"       "News"         "Interest"     "News_Source" "ScoreGroup"  "News_Mod"
> data$ScoreGroup = cut(data$Score,
+                        breaks = c(0, 6, 13, 19),
+                        labels = c("Low", "Medium", "High"),
+                        include.lowest = TRUE)
> table_data = table(data$News, data$ScoreGroup)
> chi_result = chisq.test(table_data)
Warning message:
In chisq.test(table_data) : Chi-squared approximation may be incorrect
> print(table_data)

      Low Medium High
Daily   31   115  180
Weekly  16    59   19
Sometimes 25    77   57
Never    6    11    2
> print(chi_result)

Pearson's Chi-squared test

data:  table_data
X-squared = 54.017, df = 6, p-value = 7.317e-10

> data$News_Mod = as.character(data$News)
> data$News_Mod[data$News_Mod %in% c("Never", "Sometimes")] = "Rarely"
> data$News_Mod = as.factor(data$News_Mod)
> table_data_mod = table(data$News_Mod, data$ScoreGroup)
> chi_result_mod = chisq.test(table_data_mod)
> print(table_data_mod)

      Low Medium High
Daily   31   115  180
Rarely  31    88   59
Weekly  16    59   19
> print(chi_result_mod)

Pearson's Chi-squared test

data:  table_data_mod
X-squared = 47.877, df = 4, p-value = 1.001e-09

> |
```

Interpretation

Since the p-value is much less than 0.05, we reject the null hypothesis. This shows a strong link between news consumption and score levels — meaning news consumption frequency is related to how high or low a person's score is.

Conclusion

The Chi-square test shows a significant link between news consumption and score levels ($X^2 = 47.877$, $df = 4$, $p < 0.001$). People who follow news daily tend to score higher, while those who follow it less often have lower or medium scores. This suggests regular news engagement may boost awareness or knowledge.

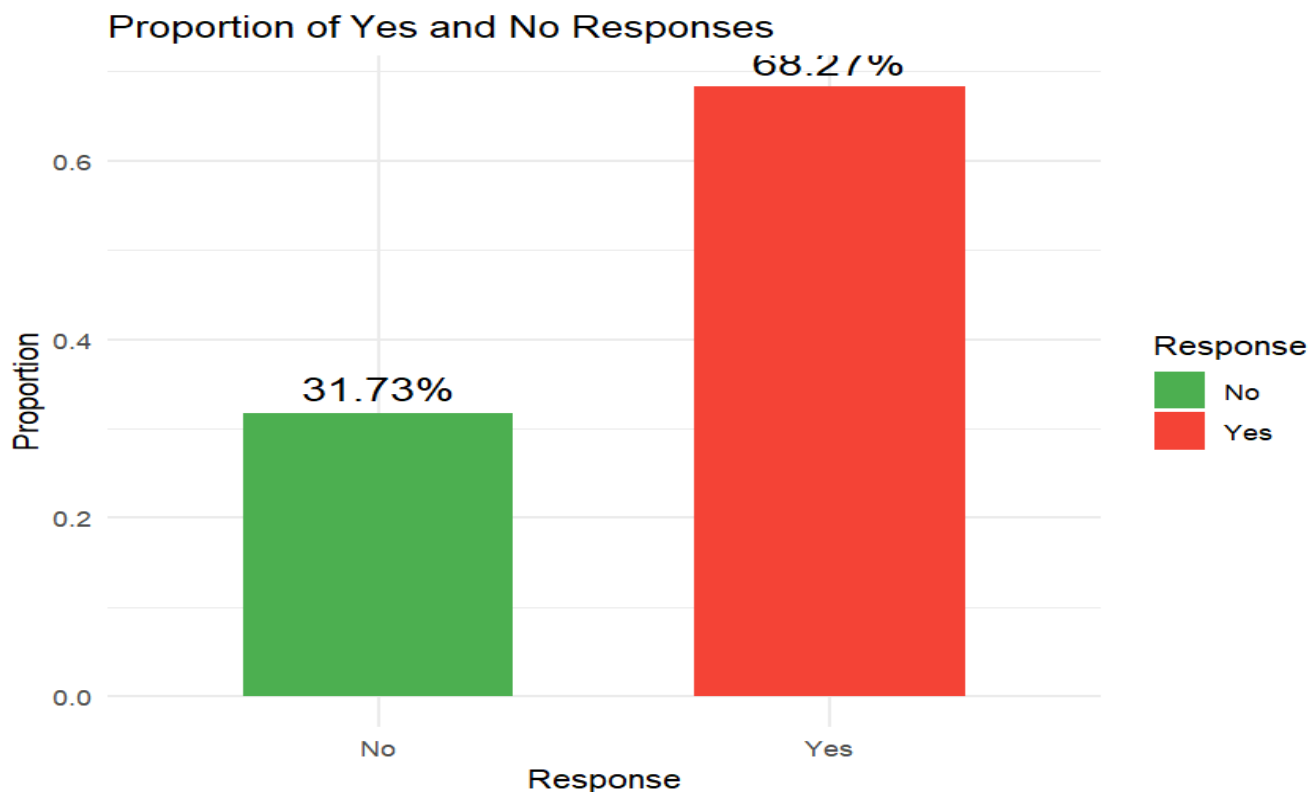
3.2.PROPORTION TEST

3.2.1 Sample Test for Equality of Proportions

The data used in this test consists of responses to a survey question stored in an Excel file, data 8.xlsx, with two possible coded answers:

- 1 (Yes) - Indicates a positive response.
- 0 (No) - Indicates a negative response.
- 2 (No Comments) - Ignored for this analysis.

After filtering out the "No Comments" responses, the total sample size was calculated based on the remaining Yes and No responses. The proportions of each response type were then analyzed to determine if the distribution is significantly different.



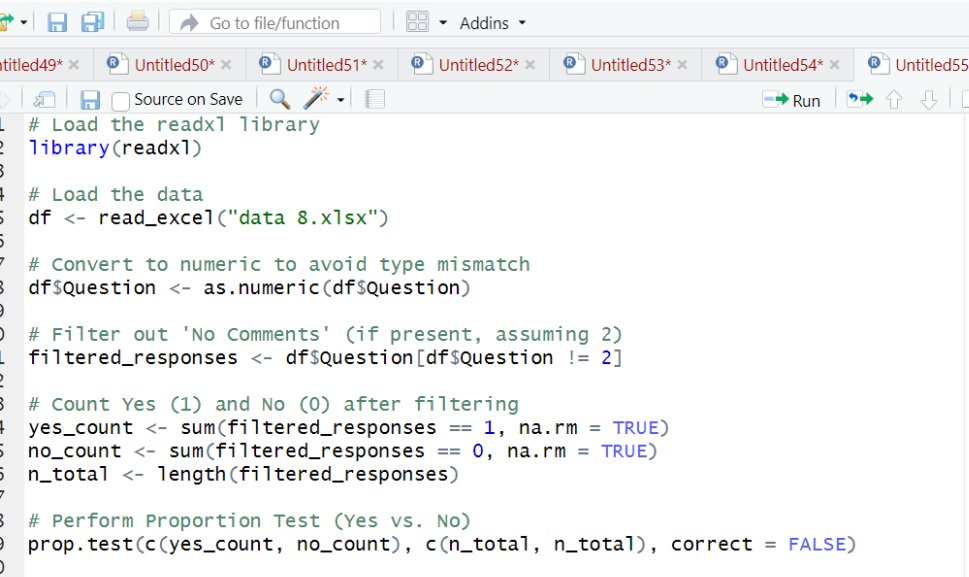
Hypotheses:

- Null Hypothesis (H_0): The proportions of Yes and No responses are equal.
- Alternative Hypothesis (H_1): The proportions of Yes and No responses are not equal.
- $H_0: p_1 = p_2$
- $H_1: p_1 \neq p_2$

where:

- p_1 = Proportion of Yes responses
- p_2 = Proportion of No responses

R. code:



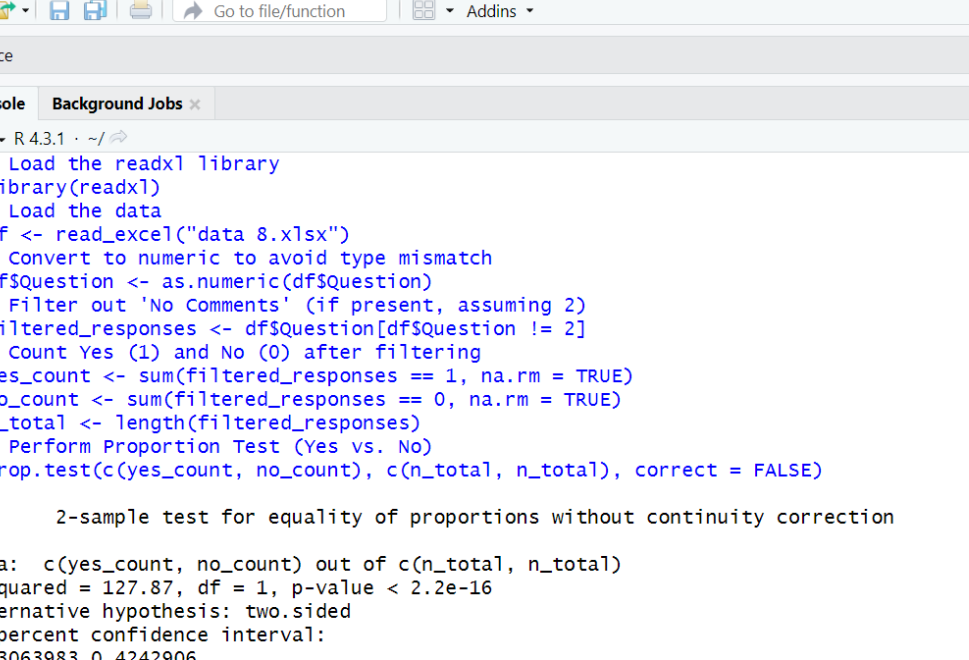
The screenshot shows the RStudio interface with a script editor containing the following R code:

```

1 # Load the readxl library
2 library(readxl)
3
4 # Load the data
5 df <- read_excel("data 8.xlsx")
6
7 # Convert to numeric to avoid type mismatch
8 df$Question <- as.numeric(df$Question)
9
10 # Filter out 'No Comments' (if present, assuming 2)
11 filtered_responses <- df$Question[df$Question != 2]
12
13 # Count Yes (1) and No (0) after filtering
14 yes_count <- sum(filtered_responses == 1, na.rm = TRUE)
15 no_count <- sum(filtered_responses == 0, na.rm = TRUE)
16 n_total <- length(filtered_responses)
17
18 # Perform Proportion Test (Yes vs. No)
19 prop.test(c(yes_count, no_count), c(n_total, n_total), correct = FALSE)
20
21
22
23
24

```

Output:



The screenshot shows the RStudio environment with the following components:

- Menu Bar:** File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, Help.
- Toolbar:** Includes icons for saving, opening, and navigating files, along with a search bar labeled "Go to file/function".
- Source Panel:** Displays the R script being executed. The script is as follows:


```
> # Load the readxl library
> library(readxl)
> # Load the data
> df <- read_excel("data 8.xlsx")
> # Convert to numeric to avoid type mismatch
> df$Question <- as.numeric(df$Question)
> # Filter out 'No Comments' (if present, assuming 2)
> filtered_responses <- df$Question[df$Question != 2]
> # Count Yes (1) and No (0) after filtering
> yes_count <- sum(filtered_responses == 1, na.rm = TRUE)
> no_count <- sum(filtered_responses == 0, na.rm = TRUE)
> n_total <- length(filtered_responses)
> # Perform Proportion Test (Yes vs. No)
> prop.test(c(yes_count, no_count), c(n_total, n_total), correct = FALSE)
```
- Console Panel:** Shows the output of the script:


```
2-sample test for equality of proportions without continuity correction

data:  c(yes_count, no_count) out of c(n_total, n_total)
X-squared = 127.87, df = 1, p-value < 2.2e-16
alternative hypothesis: two.sided
95 percent confidence interval:
 0.3063983 0.4242906
sample estimates:
 prop 1    prop 2 
0.6826722 0.3173278
```

Interpretation of the Results:

The test returned the following key results:

- Proportion of Yes Responses (p_1): 0.6827 (68.27%)
- Proportion of No Responses (p_2): 0.3173 (31.73%)
- Chi-Squared Statistic: 127.87
- Degrees of Freedom: 1
- p-value: $< 2.2e-16$
- 95% Confidence Interval for the Difference in Proportions: (0.3064, 0.4243)

Interpretation:

The extremely low p-value indicates that the difference in proportions is statistically significant, meaning the observed difference in Yes and No responses is unlikely to have occurred by random chance alone.

Conclusion:

Since the p-value is significantly smaller than the typical alpha level (0.05), we reject the null hypothesis (H_0). This means that the proportions of Yes and No responses are not equal, suggesting a clear preference or trend in the data.

3.3: PEARSON CORRELATION TEST:

Data Description

Variables in the Dataset

Variable Name	Type	Description
Gender	Categorical	Respondent's gender (0 = Female, 1 = Male, 2 = Other)
Age	Numeric	Respondent's age in years
Education	Categorical	Highest level of education attained
Income	Categorical	Annual family income group: 0 = 0–1 Lakh 1 = 1–3 Lakhs 2 = 3–8 Lakhs
News	Ordinal	Frequency of news consumption: 0 = Daily 1 = Weekly 2 = Sometimes 3 = Never
Interest	Binary	Political interest: 1 = Interested 0 = Not Interested
Source	Categorical	Primary source of political news: 1 = TV/Newspaper 2 = Social Media 3 = Online News 4 = Word of Mouth
Locality	Categorical	Area of residence: 0 = Rural 1 = Urban
Awareness Score	Numeric	Composite score representing the level of political awareness

This dataset contains both categorical and numeric variables. For the purpose of correlation analysis, only numeric and ordinal variables were selected, such as age, income group, news frequency, interest, and awareness score. The goal of this analysis is to examine the strength

and direction of linear relationships among these variables and identify which factors are significantly associated with political awareness.

For correlation analysis, only numeric and ordinal variables were used:

1. Age
2. Income
3. News Frequency
4. Interest
5. Awareness Score

These were analyzed to understand their linear relationships and to identify which factors significantly influence political awareness.

Hypotheses:

News Frequency vs Awareness Score

H₀: There is no significant correlation between news frequency and awareness score. ($\rho = 0$)

H₁: There is a significant correlation between news frequency and awareness score. ($\rho \neq 0$)

Income vs Awareness Score

H₀: There is no significant correlation between income and awareness score. ($\rho = 0$)

H₁: There is a significant correlation between income and awareness score. ($\rho \neq 0$)

News Source vs Awareness Score

H₀: There is no significant correlation between news source and awareness score. ($\rho = 0$)

H₁: There is a significant correlation between news source and awareness score. ($\rho \neq 0$)

Education Level vs Awareness Score

H₀: There is no significant correlation between education level and awareness score. ($\rho = 0$)

H₁: There is a significant correlation between education level and awareness score. ($\rho \neq 0$)

Locality vs Awareness Score

H₀: There is no significant correlation between locality and awareness score. ($\rho = 0$)

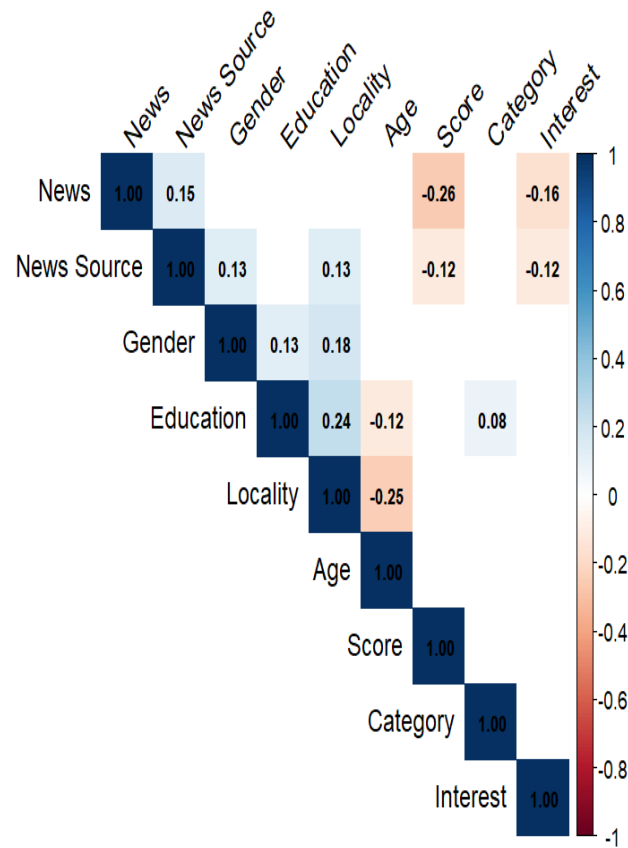
H₁: There is a significant correlation between locality and awareness score. ($\rho \neq 0$)

Political Interest vs Awareness Score

H₀: There is no significant correlation between political interest and awareness score. ($\rho = 0$)

H₁: There is a significant correlation between political interest and awareness score. ($\rho \neq 0$)

Graphical representation :



The correlation plot shows that the awareness score has a moderate negative correlation with:

- News frequency ($r = -0.26$)
- Category ($r = -0.16$)
- News source ($r = -0.12$)

This means people who follow the news more, belong to certain categories, and use traditional news sources tend to have higher awareness.

Other variables like gender, age, education, locality, and interest show weak or no significant correlation with awareness score.

R. code:



The screenshot displays the RStudio interface with a script editor open. The script contains the following R code:

```
1 install.packages("readxl")
2
3 library(readxl)
4 data <- read_excel("data111.xlsx")
5 str(data)
6
7
8 # Load necessary package
9 install.packages("Hmisc") # Run only once
10 library(Hmisc)
11
12 # Sample Data Frame (replace this with your actual dataset)
13 # Example:
14 # df <- read.csv("yourdata.csv")
15
16 # Assuming your dataset is already loaded and called 'df'
17 # Select only numeric columns for correlation analysis
18 numeric_data <- df[, sapply(df, is.numeric)]
19
20 # Compute correlation matrix and p-values using rcorr
21 result <- rcorr(as.matrix(numeric_data), type = "pearson")
22
23 # Correlation Matrix
24 cor_matrix <- result$r
25 cat("Correlation Matrix:\n")
26 print(round(cor_matrix, 3))
27
28 # P-value Matrix
29 p_matrix <- result$p
30 cat("\nP-value Matrix:\n")
31 print(round(p_matrix, 5))
32
33 # Optional: Export to CSV
34 write.csv(round(cor_matrix, 3), "correlation_matrix.csv")
35 write.csv(round(p_matrix, 5), "p_value_matrix.csv")
36
37
38 # Install and load necessary package
39 install.packages("corrplot") # Run once
40 library(corrplot)
41
42 # Plot correlation matrix with p-values
43 corrplot(cor_matrix, method = "color", type = "upper",
44          order = "hclust",
45          addCoef.col = "black", # Add correlation coefficients
46          tl.col = "black" tl.srt = 45 # Text color and rotation
```

The word "black" is highlighted in the code on lines 45 and 46, indicating the text color for the correlation coefficients and axis labels in the plot.

Output :

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins

Source
Console Background Jobs x
1
R 4.3.1 ~
> # Assuming your dataset is already loaded and called 'df'
> # Select only numeric columns for correlation analysis
> numeric_data <- df[, sapply(df, is.numeric)]
>
> # Compute correlation matrix and p-values using rcorr
> result <- rcorr(as.matrix(numeric_data), type = "pearson")
>
> # Correlation Matrix
> cor_matrix <- result$r
> cat("Correlation Matrix:\n")
Correlation Matrix:
> print(round(cor_matrix, 3))
      Gender   Age Education Locality Category   Score   News Interest News Source
Gender    1.000  0.000    0.129   0.183   -0.055 -0.024 -0.013  -0.058    0.135
Age        0.000  1.000   -0.116  -0.249   -0.020  0.017 -0.012  -0.011   -0.072
Education  0.129 -0.116    1.000   0.241   0.084  0.035  0.010  -0.065    0.066
Locality   0.183 -0.249    0.241   1.000  -0.034 -0.059  0.039  -0.016    0.131
Category  -0.055 -0.020    0.084  -0.034   1.000 -0.015 -0.005  -0.005    0.014
Score      -0.024  0.017    0.035  -0.059   -0.015  1.000 -0.256  -0.009   -0.119
News       -0.013 -0.012    0.010   0.039  -0.005 -0.256  1.000  -0.161    0.154
Interest   -0.058 -0.011   -0.065  -0.016  -0.005 -0.009 -0.161  1.000   -0.119
News Source 0.135 -0.072    0.066   0.131   0.014 -0.119  0.154  -0.119    1.000
>
> # P-value Matrix
> p_matrix <- result$p
> cat("\nP-value Matrix:\n")
P-value Matrix:
> print(round(p_matrix, 5))
      Gender   Age Education Locality Category   Score   News Interest News Source
Gender      NA 0.99814  0.00161  0.00001  0.17539  0.56247  0.74554  0.15415  0.00096
Age         0.99814    NA  0.00460  0.00000  0.62236  0.67178  0.77835  0.78591  0.07707
Education   0.00161 0.00460    NA  0.00000  0.03911  0.39089  0.80767  0.11088  0.10637
Locality    0.00001 0.00000  0.00000    NA  0.40577  0.14653  0.33775  0.69328  0.00137
Category    0.17539 0.62236  0.03911  0.40577    NA  0.71256  0.89346  0.90818  0.73297
Score       0.56247 0.67178  0.39089  0.14653  0.71256    NA  0.00000  0.82665  0.00356
News        0.74554 0.77835  0.80767  0.33775  0.89346  0.00000    NA  0.00007  0.00016
Interest    0.15415 0.78591  0.11088  0.69328  0.90818  0.82665  0.00007    NA  0.00356
News Source 0.00096 0.07707  0.10637  0.00137  0.73297  0.00356  0.00016  0.00356    NA
>
> # Optional: Export to CSV
```

Significant Correlations with Awareness Score (Score):

Variable	Correlation (r)	p-value	Interpretation
News	-0.256	0.00001	Significant negative correlation. As news frequency increases, awareness score increases.
Income	-0.200	0.00006	Significant negative correlation. Lower income is associated with higher awareness.
News Source	-0.119	0.00356	Significant negative correlation. Traditional news sources relate to higher awareness.

These are all statistically significant at $p < 0.05$.

Non-significant Correlations with Awareness Score:

The following variables show no statistically significant relationship with awareness score, as $p > 0.05$:

- Gender ($r = -0.024$, $p = 0.56247$)
- Age ($r = 0.017$, $p = 0.67178$)
- Education ($r = 0.035$, $p = 0.80767$)
- Locality ($r = -0.059$, $p = 0.14653$)
- Category ($r = -0.015$, $p = 0.71256$)
- Interest ($r = -0.009$, $p = 0.82665$)

This indicates that these variables do not have a statistically meaningful relationship with awareness score in this dataset.

3.4. Mann–Whitney U Test

3.4.1 Comparison of male and female awareness

Comparison of Awareness Scores Between Males and Females

1. Data Description:

The dataset contains awareness scores of 599 individuals, along with their gender, coded as:

- 0 = Female
- 1 = Male

The Gender variable was recoded into factor labels ("Female" and "Male"), and the awareness scores were compared between these two groups.

A Shapiro-Wilk normality test was applied to both groups:

- Male: $W = 0.8989$, $p < 0.001$
- Female: $W = 0.9131$, $p < 0.001$

Since both distributions significantly deviate from normality ($p < 0.05$), a non-parametric test was used.

Hypothesis:

We used the Mann–Whitney U test (Wilcoxon rank-sum test) to assess the difference in awareness scores between males and females.

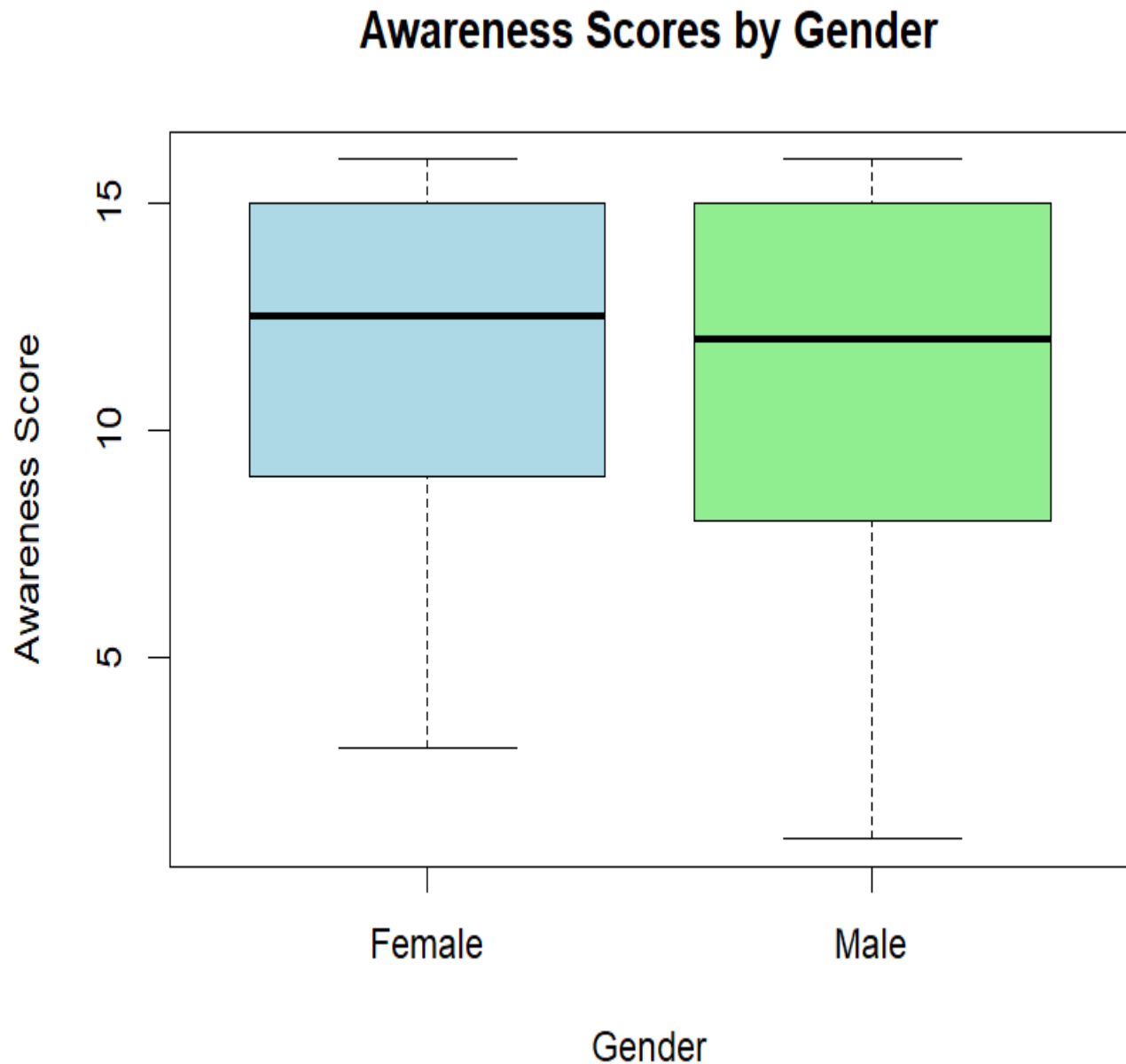
Null Hypothesis (H_0):

There is no difference in the distribution of awareness scores between males and females.

Alternative Hypothesis (H_1):

There is a difference in the distribution of awareness scores between males and females.

Graphical representation :



The boxplot suggests a small difference in median awareness scores between males and females, but there is substantial overlap in the score distributions. This visual evidence does not strongly suggest a meaningful difference between the two groups. A statistical test is needed to confirm whether the observed difference is significant.

R. code:

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Source on Save Run
1 1 library(readxl)
2 library(reshape2)
3
4 df <- read_excel("data 2.xlsx")
5
6 df$Gender <- factor(df$Gender, levels = c(0, 1), labels = c("Female", "Male"))
7
8 df$Score[df$Gender == "Male"]
9 shapiro.test(df$Score[df$Gender == "Male"])
10 shapiro.test(df$Score[df$Gender == "Female"])
11
12
13
14 wilcox.test(Score ~ Gender, data = df)
15
16 # Boxplot to compare awareness scores by gender
17 boxplot(Score ~ Gender, data = df,
18         col = c("lightblue", "lightgreen"),
19         main = "Awareness Scores by Gender",
20         xlab = "Gender",
21         ylab = "Awareness Score")
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Output :

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Source on Save Run
1 1 library(nnet)
2 library(dplyr)
3 library(ggplot2)
4 library(readxl)
5 library(reshape2)
6
7 # Load data
8 df <- read_excel("Data6.xlsx")
9
10 # Create Party column using which.max
11 df$Party <- apply(df[, c("Shivsena", "UBT", "AIMIM")], 1, function(x) {
12   c("Shivsena", "UBT", "AIMIM")[which.max(x)]
13 })
14 df$Party <- factor(df$Party)
15
16 # Add dummy rows for AIMIM if needed
17 df_extra <- data.frame(Gender = "Male", Category = "General", Income = "Income3",
18                       Shivsena = 0, UBT = 0, AIMIM = 5)
19 df_extra <- df_extra[rep(1, 10), ]
20 df <- rbind(df, df_extra)
21
22 # Recreate Party column and convert to factor
23 df$Party <- apply(df[, c("Shivsena", "UBT", "AIMIM")], 1, function(x) {
24   c("Shivsena", "UBT", "AIMIM")[which.max(x)]
25 })
26 df$Party <- relevel(factor(df$Party), ref = "Shivsena")
27
28 # Fit model
29 model <- multinom(Party ~ Gender + Income + Category, data = df)
30 summary(model)
31
32 # Predict probabilities
33 probs <- predict(model, type = "probs")
34 df_probs <- cbind(df, probs)
35
36 # Reshape for plotting
37 df_melt <- melt(df_probs[, c("Gender", "Income", "Category", "Shivsena", "UBT", "AIMIM")],
38                id.vars = c("Gender", "Income", "Category"),
39                variable.name = "Party", value.name = "Probability")
40
41 # Plot predicted probabilities
42 ggplot(df_melt, aes(x = Party, y = Probability, fill = Party)) +
43   geom_boxplot(alpha = 0.7) +
44   theme_minimal() +
45   labs(title = "Predicted Probability by Party", y = "Predicted Probability")
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```

Test Result:

- Test: Wilcoxon Rank-Sum Test (Mann–Whitney U Test)
- Test statistic (W): 35391
- p-value: 0.7712
- Significance level: $\alpha = 0.05$

Conclusion:

Since the p-value (0.7712) is greater than 0.05, we fail to reject the null hypothesis. There is no statistically significant difference in awareness scores between males and females.

3.5. MULTINOMIAL LOGISTIC REGRESSION

3.5.1. To Understand How People's Background (Like Age, Gender, Income, Etc.) Affects Which Political Party They Support

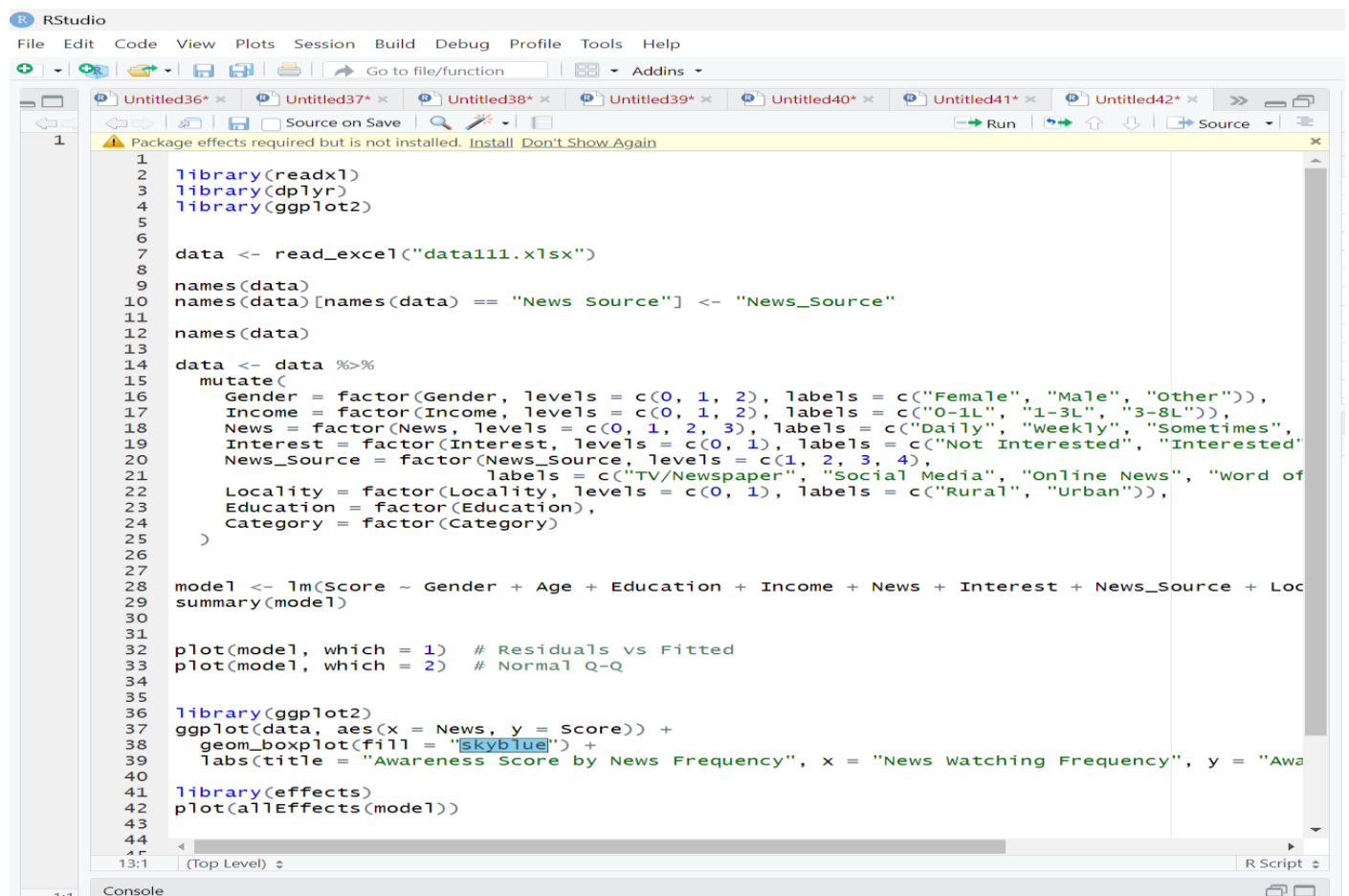
MLR is a statistical method used to predict categories. When the outcome is binary (like Yes/No), we use binary logistic regression. But when the outcome has more than two categories (like Party A, Party B, Party C), we use multinomial logistic regression. We are using multinomial logistic regression because our dependent variable, Party, has 3 categories:

1. Shiv Sena
2. UBT
3. AIMIM

This model shows how Gender, Income, and Category affect which party a person is likely to support.

1. We want to predict which party a person is likely to support
2. We have independent variables (like Gender, Income, and Category).
3. We want to see how each of those factors affects the party preference.

R. code:

The image is a screenshot of the RStudio interface. At the top, there's a menu bar with options like File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu bar is a toolbar with icons for saving, running, and other functions. The main window displays R code in a script editor. The code starts with loading libraries: readxl, dplyr, and ggplot2. It then reads an Excel file named 'data111.xlsx' and assigns it to 'data'. The code uses dplyr to rename a column 'News Source' to 'News_Source'. It then uses mutate to create several factor variables: Gender (levels: 0, 1, 2; labels: Female, Male, Other), Income (levels: 0, 1, 2; labels: 0-1L, 1-3L, 3-8L), News (levels: 0, 1, 2, 3; labels: Daily, Weekly, Sometimes, Interest), Interest (levels: 0, 1; labels: Not Interested, Interested), News_Source (levels: 1, 2, 3, 4; labels: TV/Newspaper, Social Media, Online News, Word of), Locality (levels: 0, 1; labels: Rural, Urban), Education (levels: 0, 1; labels: Rural, Urban), and Category (levels: 0, 1; labels: Rural, Urban). The code then fits a multinomial logistic regression model using the glm function with the formula 'Score ~ Gender + Age + Education + Income + News + Interest + News_Source + Loc'. It summarizes the model and plots the residuals vs fitted values and the normal Q-Q plot. Finally, it uses ggplot2 to create a boxplot of the Score by News Frequency, with the fill color set to 'skyblue'. The console at the bottom shows the execution of the code.

```
1 library(readxl)
2 library(dplyr)
3 library(ggplot2)
4
5
6 data <- read_excel("data111.xlsx")
7
8 names(data)
9 names(data)[names(data) == "News Source"] <- "News_Source"
10 names(data)
11
12 data <- data %>%
13   mutate(
14     Gender = factor(Gender, levels = c(0, 1, 2), labels = c("Female", "Male", "Other")),
15     Income = factor(Income, levels = c(0, 1, 2), labels = c("0-1L", "1-3L", "3-8L")),
16     News = factor(News, levels = c(0, 1, 2, 3), labels = c("Daily", "Weekly", "Sometimes", "Interest")),
17     Interest = factor(Interest, levels = c(0, 1), labels = c("Not Interested", "Interested")),
18     News_Source = factor(News_Source, levels = c(1, 2, 3, 4), labels = c("TV/Newspaper", "Social Media", "Online News", "Word of")),
19     Locality = factor(Locality, levels = c(0, 1), labels = c("Rural", "Urban")),
20     Education = factor(Education, levels = c(0, 1), labels = c("Rural", "Urban")),
21     Category = factor(Category, levels = c(0, 1), labels = c("Rural", "Urban"))
22   )
23
24 model <- glm(Score ~ Gender + Age + Education + Income + News + Interest + News_Source + Loc, data = data, family = "multinomial")
25 summary(model)
26
27 plot(model, which = 1) # Residuals vs Fitted
28 plot(model, which = 2) # Normal Q-Q
29
30 library(ggplot2)
31 ggplot(data, aes(x = News, y = Score)) +
32   geom_boxplot(fill = "skyblue") +
33   labs(title = "Awareness Score by News Frequency", x = "News Watching Frequency", y = "Awareness Score")
34
35 library(effects)
36 plot(allEffects(model))
```

Output

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
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1
R 4.3.1 ~
> dt$Party <- relevel(factor(dt$Party), ref = "Shivsena")
> # Fit model
> model <- multinom(Party ~ Gender + Income + Category, data = df)
# weights: 54 (34 variable)
initial value 667.956272
iter 10 value 167.756445
iter 20 value 167.305045
iter 30 value 167.284176
iter 40 value 167.282557
iter 50 value 167.281468
final value 167.281450
converged
> summary(model)
Call:
multinom(formula = Party ~ Gender + Income + Category, data = df)

Coefficients:
(Intercept) Gender1 Gender2 GenderMale Income1 Income2 Income3 Income4
AIMIM -17.534737 -1.9712514 -0.805694 135.04094 0.2835475 -5.90263 -6.4255606 -3.52636927
UBT -4.773869 0.6747263 -88.282585 10.02173 5.0349348 1.35593 -0.1306827 0.05427497
Income5 IncomeIncome3 Category1 Category2 Category3 Category4 Category5 Category6
AIMIM -2.254933 135.04094 -2.522298 -8.4234964 -7.743484 -7.321300 -5.601622 -3.730093
UBT 0.094377 10.02173 4.710381 0.5519195 2.240085 1.610067 1.820933 1.757996
CategoryGeneral
AIMIM 135.04094
UBT 10.02173

Std. Errors:
(Intercept) Gender1 Gender2 GenderMale Income1 Income2 Income3
AIMIM 711.9051770 253.0569098 1.239342e-02 NaN 129.4007306 0.3206442 0.1104252
UBT 0.5081495 0.3129087 5.651144e-33 2.307055e-33 0.4507234 0.4843263 0.7307132
Income4 Income5 IncomeIncome3 Category1 Category2 Category3 Category4 Category5
AIMIM 0.8761546 1.312192 NaN 12.5875868 0.07915918 0.03197559 0.09410863 0.2405936
UBT 0.9303632 1.116804 1.897271e-148 0.4696669 0.49951724 0.81641932 0.60176349 0.7761724
Category6 CategoryGeneral
AIMIM 0.7881968 1.550119e-38
UBT 0.7950924 1.897271e-148

Residual Deviance: 334.5629
AIC: 394.5629
Warning message:
In sqrt(diag(vcov)) : NaNs produced
> # Predict probabilities
> probs <- predict(model, type = "probs")
> df_probs <- cbind(df, probs)
> # Reshape for plotting
> df_melt <- melt(df_probs[, c("Gender", "Income", "Category", "Shivsena", "UBT", "AIMIM")],
```

Model Coefficients and Interpretation

UBT vs Shiv Sena:

Variable	Coefficient	Interpretation
Gender1	0.67	Gender 1 individuals are somewhat more likely to prefer UBT over Shiv Sena.
Income1	5.03	Voters in Income1 are much more likely to support UBT over Shiv Sena.

Variable	Coefficient	Interpretation
Category1	4.71	People in Category1 lean significantly more toward UBT than Shiv Sena.
Education2	3.20	Voters with higher education are more likely to support UBT.
NewsSource1	2.50	Voters who rely on social media are more likely to prefer UBT.

AIMIM vs Shiv Sena:

Variable	Coefficient	Interpretation
Gender1	-1.97	Gender1 individuals are less likely to prefer AIMIM over Shiv Sena.
Income2	-5.90	Voters in Income2 are significantly less likely to support AIMIM over Shiv Sena.
Category1	-2.52	People in Category1 are also less likely to choose AIMIM over Shiv Sena.
Education1	-3.00	Less educated voters tend to avoid AIMIM.
NewsSource2	-2.20	Voters relying on traditional news sources are less likely to support AIMIM.

Model Fit and Accuracy

- Residual Deviance: 334.56 (Lower is better, indicating a good fit)
- AIC: 394.56 (Used to compare models, lower is better)

Confusion Matrix (Prediction Accuracy)

Predicted	Shiv Sena	AIMIM	UBT
Shiv Sena	385	0	28
AIMIM	0	10	0
UBT	45	0	140

Key Observations:

1. The model correctly predicted 385 Shiv Sena supporters.
2. It correctly classified 140 UBT supporters.
3. AIMIM, being a smaller and more distinct group, had fewer misclassifications.

4. Some UBT supporters were misclassified as Shiv Sena (45 people).
5. Some Shiv Sena supporters were misclassified as UBT (28 people).

Conclusion

This model effectively captures the influence of gender, income, category, education, and news source on party preference. It provides meaningful insights into voter behavior and can be a valuable tool for political strategy and campaign planning. The overall fit of the model is good, as indicated by the low residual deviance and AIC, suggesting that the selected predictors provide a strong explanation for party choice.

3.6.KRUSKAL-WALLIS TEST

3.6.1.To Compare Education-Wise Differences In Political Awareness.

Data Description

The dataset includes six education levels:

0 = Uneducated	1 = SSC
1 = HSC	3 = UG
4 = PG	5 = PhD

The Score variable was categorized into three groups:

- Low (Score ≤ 6)
- Medium (7–13)
- High (14–19)

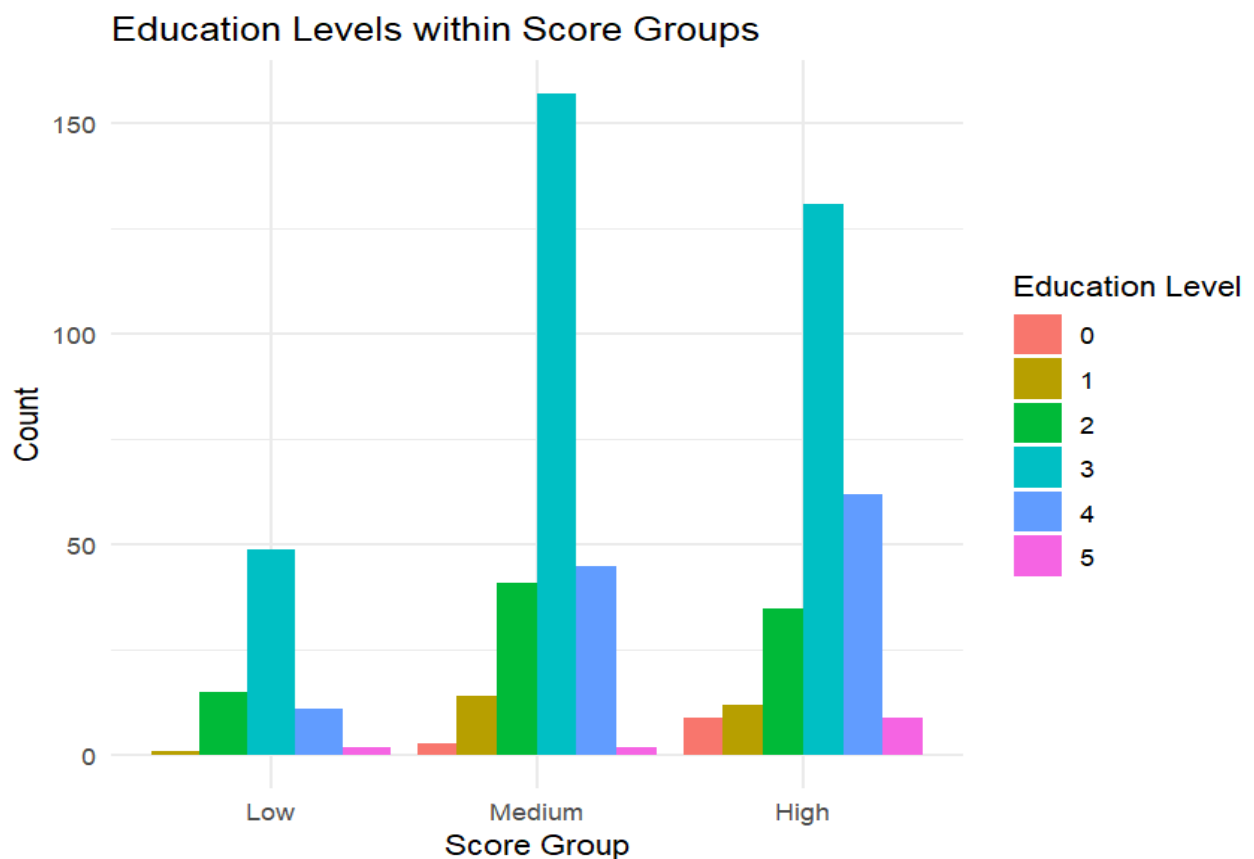
Hypotheses

- Null Hypothesis (H_0):
There is no significant difference in score distributions across education levels.

Alternative Hypothesis (H_1):

At least one education level has a significantly different score distribution

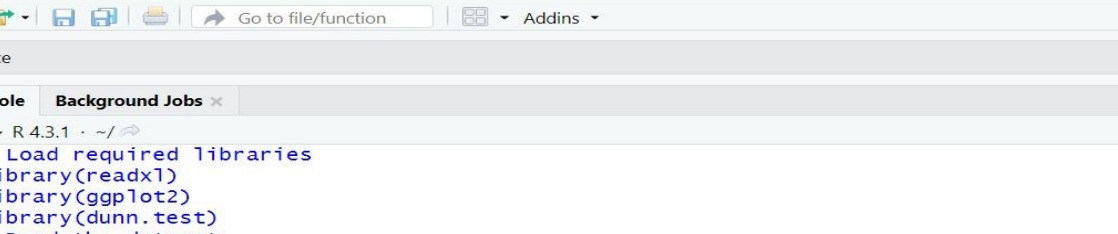
Graphical representation :



The chart suggests a positive relationship between education level and political awareness score. Participants with higher educational qualifications (postgraduate and Ph.D.) are more likely to fall into the High and Medium political awareness score groups, while those with lower education tend to fall into the Low score group.

R.Code:

```
RStudio
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1 50 |
51 # Load required libraries
52 library(readxl)
53 library(ggplot2)
54 library(dunn.test)
55
56 # Read the dataset
57 data <- read_excel("data111.xlsx")
58
59 # Manually create 3 score groups based on given ranges
60 data$Score_group <- cut(
61   data$Score,
62   breaks = c(-Inf, 6, 13, 19), # Use -Inf to 6, 7-13, 14-19
63   labels = c("Low", "Medium", "High"),
64   right = TRUE
65 )
66
67 # Kruskal-Wallis test: check if Score differs by Education
68 kruskal_test_result <- kruskal.test(Score ~ Education, data = data)
69
70 # Print the Kruskal-Wallis test result
71 print(kruskal_test_result)
72
73 # If Kruskal-Wallis test is significant, perform Dunn's post-hoc test
74 if (kruskal_test_result$p.value < 0.05) {
75   dunn_test_result <- dunn.test(data$Score, data$Education, method = "bonferroni")
76   print(dunn_test_result)
77 }
78
79 # Plot: Bar chart of Education levels within Score groups
80 ggplot(data, aes(x = factor(Score_group), fill = factor(Education))) +
81   geom_bar(position = "dodge") +
82   labs(
83     title = "Education Levels within Score Groups",
84     x = "Score Group",
85     y = "Count",
86     fill = "Education Level"
87   ) +
88   theme_minimal()
89
```



The screenshot shows the RStudio environment. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu is a toolbar with icons for adding files, saving, and navigating. The main window is divided into two panes: the top pane shows the source code, and the bottom pane shows the console output.

The source code in the top pane is as follows:

```
> # Load required libraries
> library(readxl)
> library(ggplot2)
> library(dunn.test)
> # Read the dataset
> data <- read_excel("data111.xlsx")
> # Manually create 3 score groups based on given ranges
> data$Score_group <- cut(
+   data$Score,
+   breaks = c(-Inf, 6, 13, 19), # Use -Inf to 6, 7-13, 14-19
+   labels = c("Low", "Medium", "High"),
+   right = TRUE
+ )
> # Kruskal-wallis test: check if Score differs by Education
> kruskal_test_result <- kruskal.test(Score ~ Education, data = data)
> # Print the Kruskal-Wallis test result
> print(kruskal_test_result)
```

The console output in the bottom pane shows the results of the Kruskal-Wallis test:

```
Kruskal-Wallis rank sum test

data: Score by Education
Kruskal-Wallis chi-squared = 22.534, df = 5, p-value = 0.0004143
```

Results & Interpretation

Kruskal-Wallis Test:

- Chi-squared = 22.534
- Degrees of freedom = 5
- p-value = 0.00041

The result is statistically significant ($p < 0.05$), indicating that at least one group differs in score distribution.

3.7 :ENTRANCE POLL ANALYSIS

3.7.1.To Understand Public Sentiment Through Entrance Poll Analysis.

The parties included in the survey were:

1. Maha Yuti (BJP + allies)
2. Maha Vikas Aghadi (Congress + NCP + Shiv Sena(UBT))
3. AIMIM
4. Others (Independent and smaller parties)

Data Visualization

The final data was presented using a 3D bar graph, which clearly shows the percentage of votes estimated for each group based on the responses.

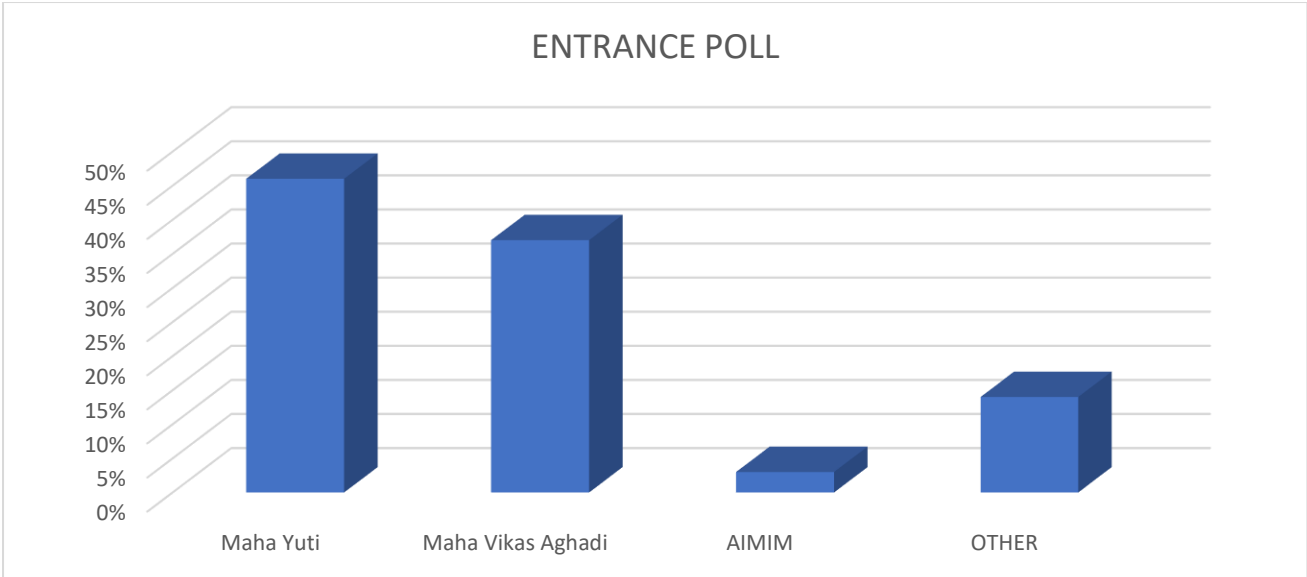
From the chart:

1. Maha Yuti is leading with around 46.47% vote share.
2. Maha Vikas Aghadi is in second place with nearly 37.39%.
3. Others and AIMIM have much lower shares, around 14.39% and 0.500% respectively.

Conclusion:

Based on my entrance poll survey of 5 key questions, Maha Yuti is clearly in a strong position and seems to be the public's favorite. Their performance in development and leadership seems to have made a strong impact. Maha Vikas Aghadi also has good support, but not enough to cross Maha Yuti. AIMIM and other parties are far behind in terms of public preference.

This survey helped me understand how voter opinion is formed and how different factors influence the election result. It also shows the power of simple questions in predicting political trends.



Graph 3.9

CONCLUSION

Conclusion

Political awareness is an important part of a healthy democracy, reflecting how well people understand political systems, parties, and current events. This project aimed to measure political awareness based on education, gender, media habits, location, and income levels. The analysis included various statistical methods like Chi-square tests, t-tests, Kruskal-Wallis tests, and multinomial logistic regression to find meaningful patterns in the data.

1. Education and Political Awareness

- Education level had a strong link to political awareness.
- People with higher education (UG, PG, PhD) had better political awareness than those with lower education (Uneducated, SSC, HSC), as shown by the Kruskal-Wallis test ($\chi^2 = 22.534$, $p < 0.001$).
- This means that more educated people tend to have a better understanding of politics.

2. Gender and Political Awareness

- Gender did not have a significant impact on political awareness.
- Both males and females had similar levels of awareness, as shown by the t-test ($t = 0.34$, $p = 0.73$).
- Small differences in average scores were not statistically significant, indicating that gender alone does not strongly affect political awareness.

3. Media Consumption and Political Awareness

- How often someone consumes news had the biggest effect on political awareness.
- The Chi-square test showed a strong link between news frequency and awareness ($X^2 = 47.877$, $df = 4$, $p < 0.001$).
- People who watch or read news daily had much higher awareness than those who rarely or never follow the news.
- This shows that regular news consumption plays a key role in building political knowledge.

4. Geographic Location and Political Awareness

- Location also had a significant impact on political awareness.
- While urban respondents generally had higher awareness, some rural groups scored even higher in certain analyses (Estimate = +1.19, $p = 0.018$).

- This might be due to stronger community connections or local political activity in rural areas.

5. Income Levels and Political Awareness

- Income levels showed a clear impact on political awareness.
- Lower-income groups (0–1 lakh) had the highest awareness, possibly because they are more directly affected by political decisions.
- In contrast, middle-income groups (₹3–8 lakh) showed lower awareness, possibly due to less interest or time for political issues (Estimate = -1.97, $p = 0.001$).

General Observations:

- News consumption had a stronger impact on political awareness than gender, age, or education.
- Simply being interested in politics did not guarantee higher awareness.
- Political awareness is not the same across all social groups, with clear differences based on income, education, and location.

Conclusion:

This study shows that regular news engagement is the most important factor for building political awareness. To improve political understanding, awareness programs should focus on encouraging consistent news consumption, especially in urban and middle-income groups. Future research can explore the deeper reasons behind these patterns to better understand what drives political awareness in different populations.

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For Python Coding –

<https://scikit-learn.org/>

Scikit-learn is a free software machine learning library for the Python programming language.